Name:

GCSE COMPUTER SCIENCE

COMP1 Revision Booklet

Your Essential Exam Revision Guide

Welcome to your revision guide. It will help guide you through the key ideas you *really* need to know. You'll find clear explanations, helpful reminders, and practice questions to train your brain step by step.

Use this booklet little and often to organise your thinking, grow your confidence, and walk calmly into your exams knowing you're ready.

Paper 1 Exam Date:
Summer 2027 - check your exam timetable

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Summary of Paper 1

PAPER OVERVIEW

Paper code:	J277/01
Total marks:	80 marks (50% of GCSE)
Duration:	1 hour 30 minutes
Calculator:	Not allowed

TIMING ADVICE

- ✓ 1 mark = 1 minute
- ✓ 4 marks: 4–5 mins
- ✓ 6 marks: 7–9 mins
- ✓ Calculations: Show working!

6 MARK QUESTION TIP: TIER

- **T** – Technology
- **I** – Issue
- **E** – Effect
- **R** – Result

CALCULATION QUESTIONS

- ✓ Show your working
- ✓ Label units (e.g. KB, MB)
- ✓ Check your steps

COMMON MISTAKES

Listing, *not explaining*	No units	Naming wrong law	Skipping steps

TYPES OF QUESTIONS YOU CAN EXPECT

Multiple choice
Questions with four options – pick the correct answer.

Define
Write a short definition of a key term using correct technical voc.

Explain
Give a clear and detailed explanation of how or why something works or happens.

Compare
Explain the similarities and differences between two things, using key terms.

Calculate
Solve a problem and show your working step-by-step, giving the correct units for your final answer.

Discuss/Evaluate
Review a topic, giving both sides of an argument and reaching a balanced conclusion.

Match/Link questions
Draw lines to link terms, laws or hardware to correct definitions, purposes, or functions.

Label diagrams
Add labels to diagrams (CPU, networks, etc.)



Match/Link questions
Draw lines to link terms, laws or hardware to correct definitions, purposes, or functions.

1 mark = 1 minute. *Answer everything!*

1.1 Systems Architecture

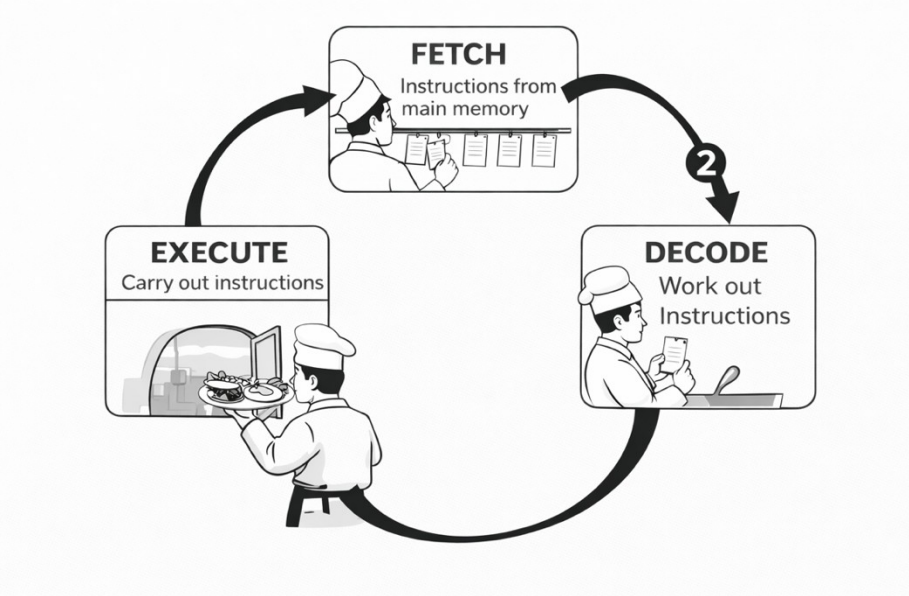
Keyword	Definition
Accumulator	Stores the result of arithmetic operations
Program Counter	Stores the address of the next instruction to be fetched from memory. Increments in each Fetch-Decode-Execute cycle.
Control Unit	Sends signals to synchronise/control/coordinates the processor/hardware/FDE cycle/processes/flow of data.
Memory Address Register	Stores the address of the data to be fetched from, or the address where the data is to be stored.
Arithmetic Logic Unit	Performs the mathematical and logical calculations
Memory Data Register	It contains the instruction which has just been copied from main memory.
Clock Speed	How many instruction cycles the CPU can deal with in a second. Measured in MHz or GHz.
Cache Size	The amount of internal storage the CPU has to store instructions.
Cores	The more cores, the more instructions the CPU can process at once and therefore the faster it performs

1.1.1 Architecture of CPU

1.1.1 Architecture of the CPU	
☐ The purpose of the CPU: ○ The fetch-execute cycle ☐ Common CPU components and their function: ○ ALU (Arithmetic Logic Unit) ○ CU (Control Unit) ○ Cache ○ Registers ☐ Von Neumann architecture: ○ MAR (Memory Address Register) ○ MDR (Memory Data Register) ○ Program Counter ○ Accumulator	**Required** ✓ What actions occur at each stage of the fetch-execute cycle ✓ The role/purpose of each component and what it manages, stores, or controls during the fetch-execute cycle ✓ The purpose of each register, what it stores (data or address) ✓ The difference between storing data and an address **Not required** × Knowledge of passing of data between registers in each stage

What is the Purpose of the CPU?

The CPU is often known as the 'brain of the computer'. Its job is to process instructions and data to run programs. The CPU follows three steps in order to process data. It is known as the Fetch - Execute cycle (aka Fetch-Decode-Execute Cycle). To explain the CPU, think about the role of a head chef in a restaurant.



The Fetch–Execute Cycle

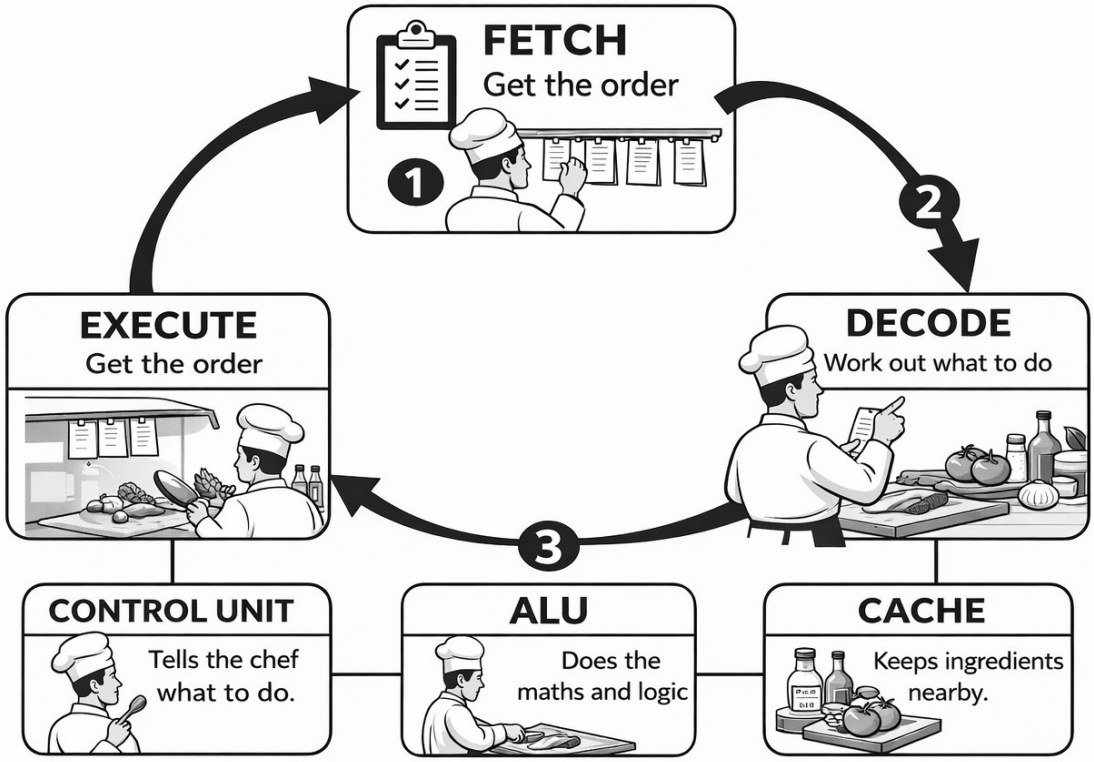
Stage	What happens	Analogy	Exam wording
Fetch	The next instruction is fetched from memory.	Chef picks up the next order.	Instruction is fetched from memory.
Decode	The instruction is worked out by the control unit.	Chef reads and understands the order.	Instruction is decoded by the control unit.
Execute	The instruction is carried out.	Chef cooks the meal.	Instruction is executed.

CPU Components

The CPU has three parts that has distinct roles in the Fetch-Decode-Execute cycle.
ALU (arithmetic and logic unit), CU (Control Unit) and Cache (fast memory of recent data and instructions)

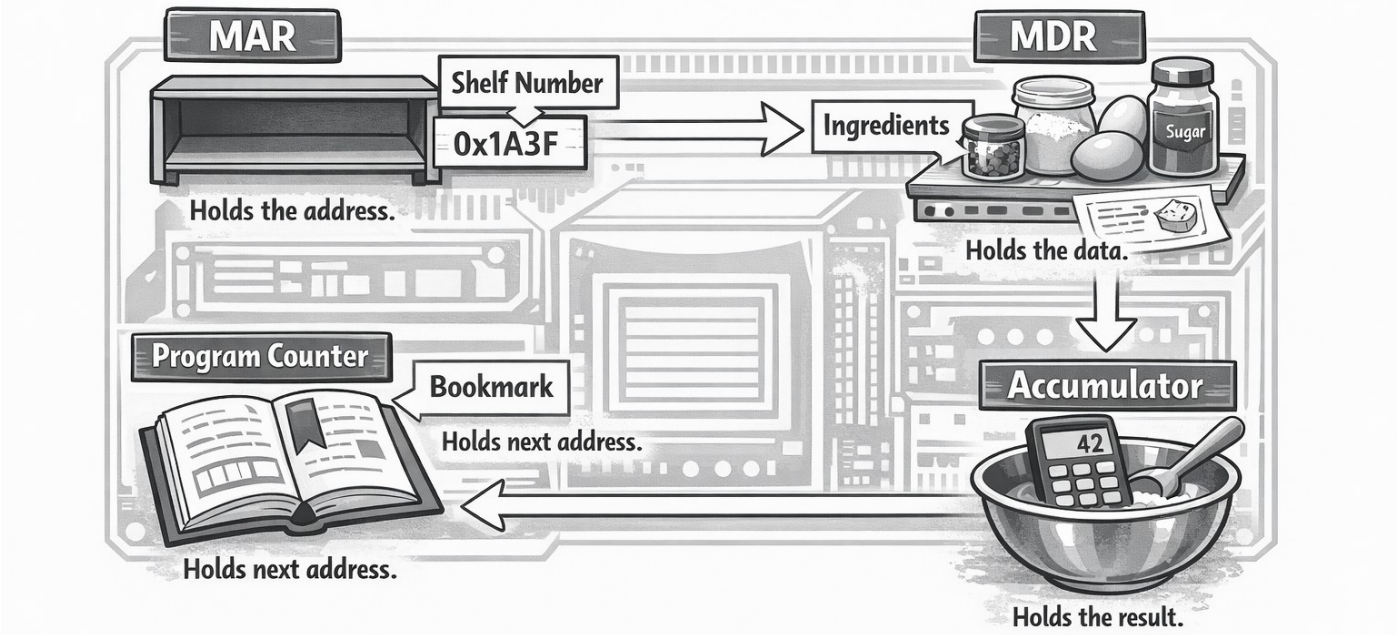
Component	Purpose	Analogy	Exam wording
ALU	Performs arithmetic and logic operations. Arithmetic - +-/*MOD DIV ^ Logic- AND OR NOT	Where the chef actually does the work (chopping, mixing and cooking), using maths to measure amounts and logic to do things in the right order.	The ALU carries out arithmetic and logical operations.
Control Unit	Controls the CPU and decodes instructions.	When a customer orders a burger**,** the chef does not cook it themselves**.** Instead, the head chef reads the order and breaks it into simple instructions.	The control unit manages and instructs the CPU.
Cache	Stores frequently used data and instructions.	Often used things like, Ingredients kept near	Cache stores frequently used data for quick access.

		the chef, like ketchup and recipes.	



Registers

Registers are very small, very fast storage spaces inside the CPU that hold data and addresses the CPU is using right now, so processing can happen quickly. There are four you need to know about.



Register	What it stores	Data or Address	Analogy	Exam wording
MAR	Memory location to be accessed.	Address	Shelf number.	The MAR stores the address of a memory location.
MDR	Data or instruction.	Data	Ingredient taken from shelf.	The MDR stores data or instructions.
Program Counter	Address of next instruction.	Address	Bookmark.	The PC stores the address of the next instruction.
Accumulator	Results of calculations.	Data	Mixing bowl.	The accumulator stores calculation results.

In a kitchen, the **address** is telling the chef which **shelf** or fridge an ingredient is on.
The **data** is the actual **ingredient** or instruction the chef takes and uses to make the meal**.**

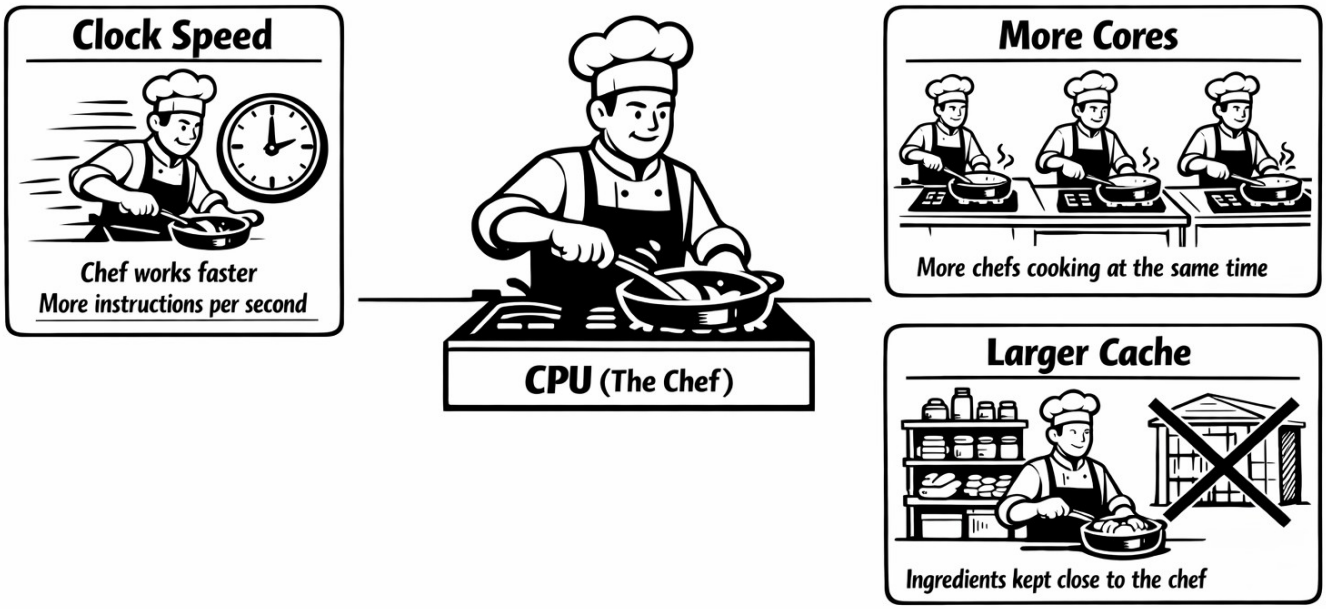
1.1.1 Exam Questions on Architecture of CPU

M	Question type	Question	Model answer	Tip to answer (student)
1	**Define**	What is the purpose of the CPU?	The CPU processes data and instructions to run programs.	One clear sentence only.
1	**Define**	Name one component of the CPU.	ALU / Control Unit / Cache.	Just name it — no explanation.
1	**Define**	What does the Program Counter store?	The address of the next instruction.	Look for the word *address*.
2	**Explain**	State two roles of the Control Unit.	It decodes instructions and controls the operation of the CPU.	Two short points = two marks.
2	**Explain**	State two things the ALU does.	It performs arithmetic operations and logical operations.	Name one maths and one logic role.
2	**Match / Link**	Match each CPU component to its function.	ALU → calculations; CU → control; Registers → fast storage.	One correct link = one mark.
3	**Explain**	Explain the Fetch–Decode–Execute cycle.	The CPU fetches the instruction from memory, decodes it using the Control Unit, and then executes it, for example by the ALU carrying out a calculation.	One sentence per stage.
3	**Explain**	Explain why registers are used inside the CPU.	Registers are very small and fast storage locations that hold data and addresses currently being used, allowing instructions to be processed quickly.	Say *fast*, *small*, *used now*.
3	**Explain**	Explain the role of cache memory in the CPU.	Cache stores frequently used data and instructions close to the CPU so they can be accessed quickly, improving performance.	Mention *frequently used* and *speed*.
3	**Label diagram**	Label the ALU, Control Unit and registers on a CPU diagram.	Correctly labelled components.	Use exact terms from the specification.
4	**Describe roles**	Describe the role of the ALU and the Control Unit.	The ALU performs arithmetic and logical operations. The Control Unit decodes instructions and controls the operation of the CPU by telling components what to do.	Two components, one role each.
4	**Describe roles**	Describe the role of the MAR and MDR.	The MAR stores the address of the memory location being accessed. The MDR stores the data or instruction fetched from memory.	MAR = address, MDR = data.
4	**Compare**	Compare the MAR and MDR.	The MAR stores addresses, whereas the MDR stores data or instructions. Both are registers used during memory access.	Use *whereas*to compare directly.
6	**Discuss / Explain**	Explain how the CPU processes instructions using the Fetch–Decode–Execute cycle.	The Program Counter stores the address of the next instruction, which is copied to the MAR. The instruction is fetched from memory and stored in the MDR. The Control Unit decodes the instruction, and during execution the ALU may perform a calculation, storing the result in the accumulator.	Follow the cycle, name registers, explain what happens.

1.1.2 Systems Architecture

1.1.2 CPU performance	
☐ How common characteristics of CPUs affect their performance: ○ Clock speed ○ Cache size ○ Number of cores	**Required** ✓ Understanding of each characteristic listed ✓ The effects of changing any of the common characteristics on system performance, either individually or in combination

CPU performance is how fast and how well a computer can do jobs, including how quickly it follows instructions and how many things it can do at the same time



A CPU can be made faster by increasing clock speed, adding more cores, or increasing cache size.

Characteristic	Purpose	Analogy	Exam wording
Clock speed measured in Hertz – FDE cycles per second	Determines how many instructions the CPU can process per second	The chef works faster, completing more cooking steps each second	Increasing clock speed allows the CPU to execute more instructions per second.
Cache size	Stores frequently used data and instructions close to the CPU to reduce access time	Ingredients are kept next to the chef instead of going to the storeroom	Increasing cache size reduces the time taken to access frequently used data.
Number of CPU cores	Allows multiple instructions to be processed at the same time	More chefs cooking different dishes at the same time	Multiple cores allow instructions to be executed in parallel.

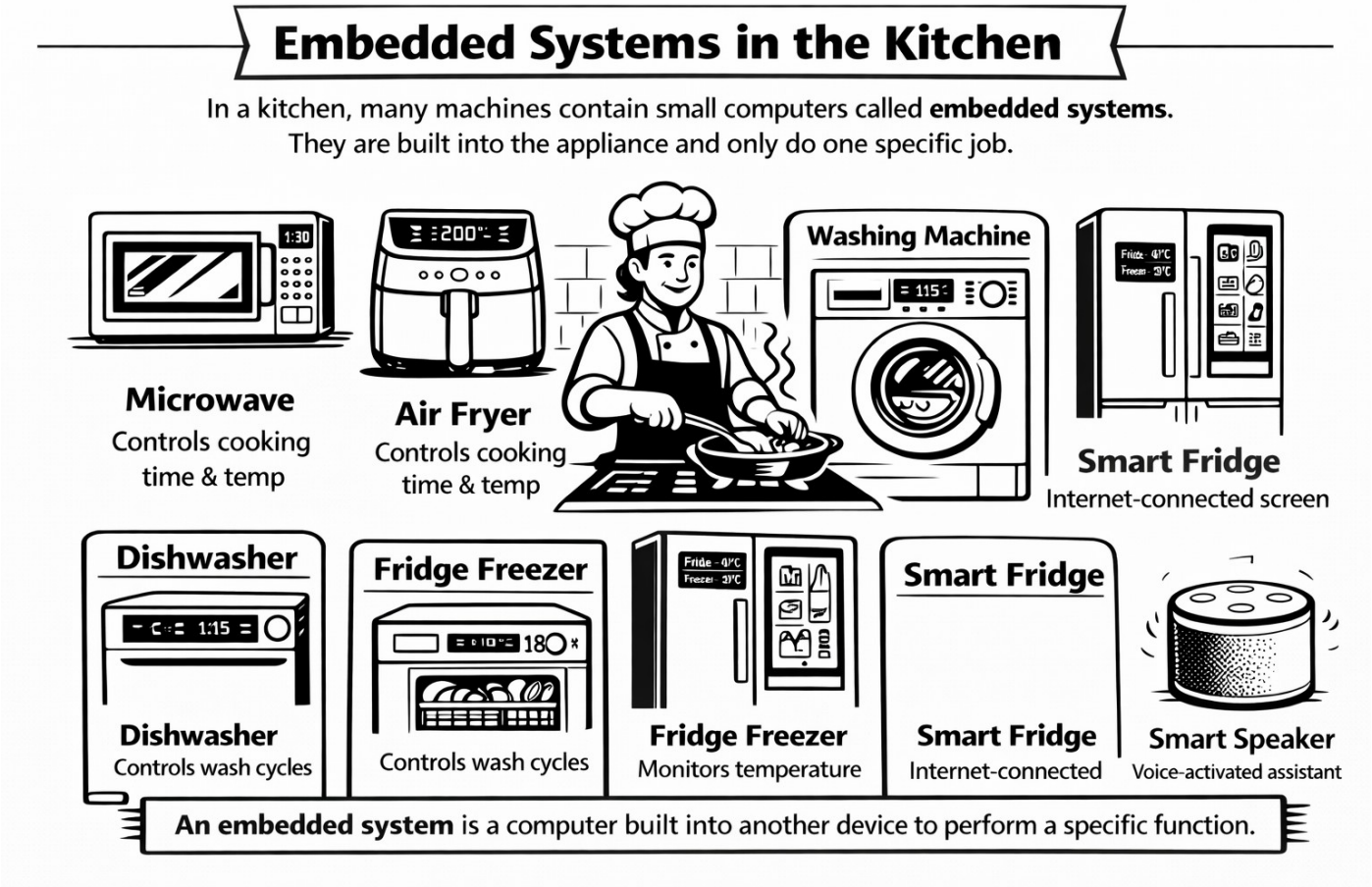
1.1.2 Exam Questions on Systems Architecture

M	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is meant by CPU performance?	CPU performance is how fast and efficiently a CPU can process instructions.	One short definition. Use the word **speed** or **efficiency**.
1	Define	Name one factor that affects CPU performance.	Clock speed / Cache size / Number of cores.	Just name one — no explanation.
1	Define	What is clock speed measured in?	Hertz (Hz).	Units question → write **Hz**.
2	Explain	State two ways increasing clock speed improves CPU performance.	It allows more instructions to be executed per second and increases the speed of the fetch–decode–execute cycle.	Two clear points — no long sentences.
2	Explain	State two benefits of having multiple CPU cores.	Multiple cores allow instructions to be processed at the same time and improve multitasking performance.	Think **parallel**and **multitasking**.
3	Explain	Explain how clock speed affects CPU performance.	Clock speed determines how many fetch–decode–execute cycles the CPU can carry out per second. A higher clock speed means more instructions can be processed each second.	Mention **FDE cycles per second**.
3	Explain	Explain the role of cache memory in improving CPU performance.	Cache stores frequently used data and instructions close to the CPU, reducing access time and increasing processing speed.	Say **frequently used** + **faster access**.
3	Explain	Explain how the number of CPU cores affects performance.	Having multiple CPU cores allows instructions to be executed in parallel, improving performance when running multiple tasks at the same time.	Use the word **parallel**.
4	Describe roles	Describe two factors that affect CPU performance.	Clock speed affects how many instructions can be processed per second, while cache size reduces the time taken to access frequently used data.	Two factors, one sentence each.
4	Describe roles	Describe how cache size and number of cores improve CPU performance.	A larger cache stores more frequently used data close to the CPU, reducing access time. Multiple cores allow several instructions to be processed at the same time.	Link each factor to **speed**.
6	Discuss / Explain	Explain how clock speed, cache size and number of cores affect CPU performance.	Clock speed determines how many fetch–decode–execute cycles occur per second. Cache size reduces the time needed to access frequently used data and instructions. The number of CPU cores allows instructions to be processed in parallel, improving multitasking and overall performance.	Structure answer by factor. One paragraph per factor = full marks.

1.1.3 Embedded Systems

1.1.3 Embedded systems	
☐ The purpose and characteristics of embedded systems ☐ Examples of embedded systems	**Required** ✓ What embedded systems are ✓ Typical characteristics of embedded systems ✓ Familiarity with a range of different embedded systems

When we think of a computer, we often think of a laptop, tablet, or PC that can run many different programs. An embedded system is different because it is a computer built into a larger device to perform one specific job.



Characteristic	What it means (simple)	Exam wording
Single purpose	Designed to do one specific job	An embedded system performs a single specific function.
Built into another device	Part of a larger machine, not a PC	The system is built into another device.
Limited user interface	Few buttons or a small screen	Embedded systems usually have a limited user interface.
Real-time operation	Responds immediately to inputs	Many embedded systems operate in real time.
Low power consumption	Uses very little electricity	Embedded systems are designed to use low power.
**Large ROM	Software stored in ROM	Embedded systems run software stored in ROM.
**Small amount of RAM	Only needs a little memory	Embedded systems have a small amount of RAM.

**See 1.2 Memory & Storage.

1.1.3 Exam Questions on Embedded Systems

M	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is an embedded system?	An embedded system is a computer built into a larger device to perform a single specific function.	Use the words **built into** and **single job**.
1	Define	Name one characteristic of an embedded system.	Single purpose / Low power / Real-time / Limited user interface.	Just name one — no explanation.
1	Define	Where is the software for an embedded system usually stored?	In ROM.	ROM = permanent storage.
2	Explain	State two characteristics of an embedded system.	Embedded systems are designed for a single purpose and are built into a larger device.	Two clear points = two marks.
2	Explain	State two reasons embedded systems use low power.	They often run continuously and are used in devices with limited power supplies such as batteries.	Think **always on** and **battery-powered**.
3	Explain	Explain what is meant by a real-time embedded system.	A real-time embedded system responds immediately to inputs within a fixed time limit to ensure correct operation.	Use **immediate** or **fixed time**.
3	Explain	Explain why embedded systems have a limited user interface.	Embedded systems perform a single function and do not need complex input or output, so only a small interface is required.	Link **single purpose** to **few buttons**.
3	Explain	Explain why embedded systems use a large ROM and a small amount of RAM.	The program is permanently stored in ROM, while only a small amount of RAM is needed to store temporary data during operation.	ROM = program, RAM = temporary data.
4	Describe roles	Describe two characteristics of embedded systems.	Embedded systems are built into a larger device and are designed to perform a single specific function.	Two characteristics, one sentence each.
4	Describe roles	Describe how real-time operation and low power consumption are important in embedded systems.	Real-time operation ensures the system responds immediately to inputs, while low power consumption allows the device to run efficiently for long periods.	Say **why** each feature matters.
6	Discuss / Explain	Explain the main characteristics of embedded systems.	Embedded systems are built into larger devices to perform a single specific function. They often operate in real time, have a limited user interface, use low power, and run software stored in ROM with only a small amount of RAM. These features make them reliable and efficient for their intended task.	Structure answer by feature. One sentence per characteristic.

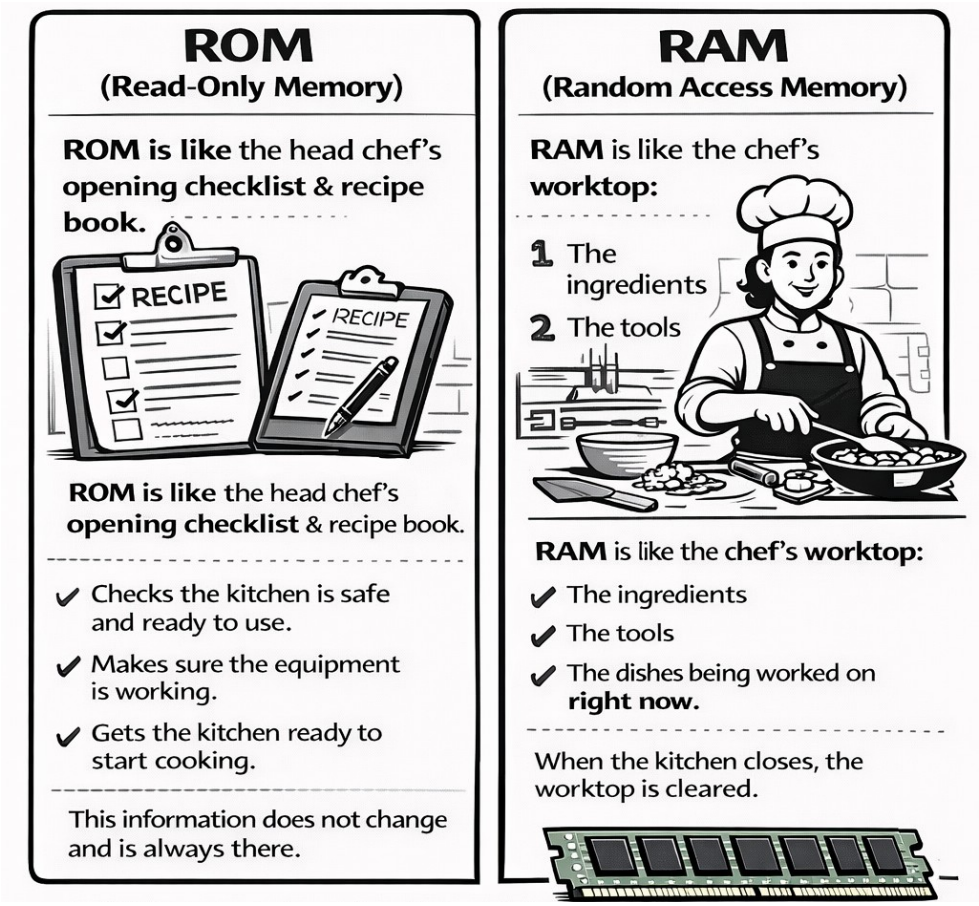
Keyword	Definition
Sample Rate	The number of times the amplitude is recorded per second.
Bit Depth	The range of sounds recorded.
Primary Storage	Stored data/instructions/software/OS that the processor needs to access.
Secondary Storage	To store data/files long-term/permanently.
RAM	Random Access Memory. Stores programs and data before they are processed by the CPU.
ROM	Read Only Memory. Data can be retrieved but not changed and the contents are not lost when the computer is turned off.
Volatile	Data stored in volatile memory is lost when the computer is turned off.
Non-Volatile	Data stored in non-volatile memory is not lost when the computer is turned off.
Virtual Memory	Space on the computer's hard drive used as primary memory (RAM).

1.2 Memory & Storage

1.2.1 Memory

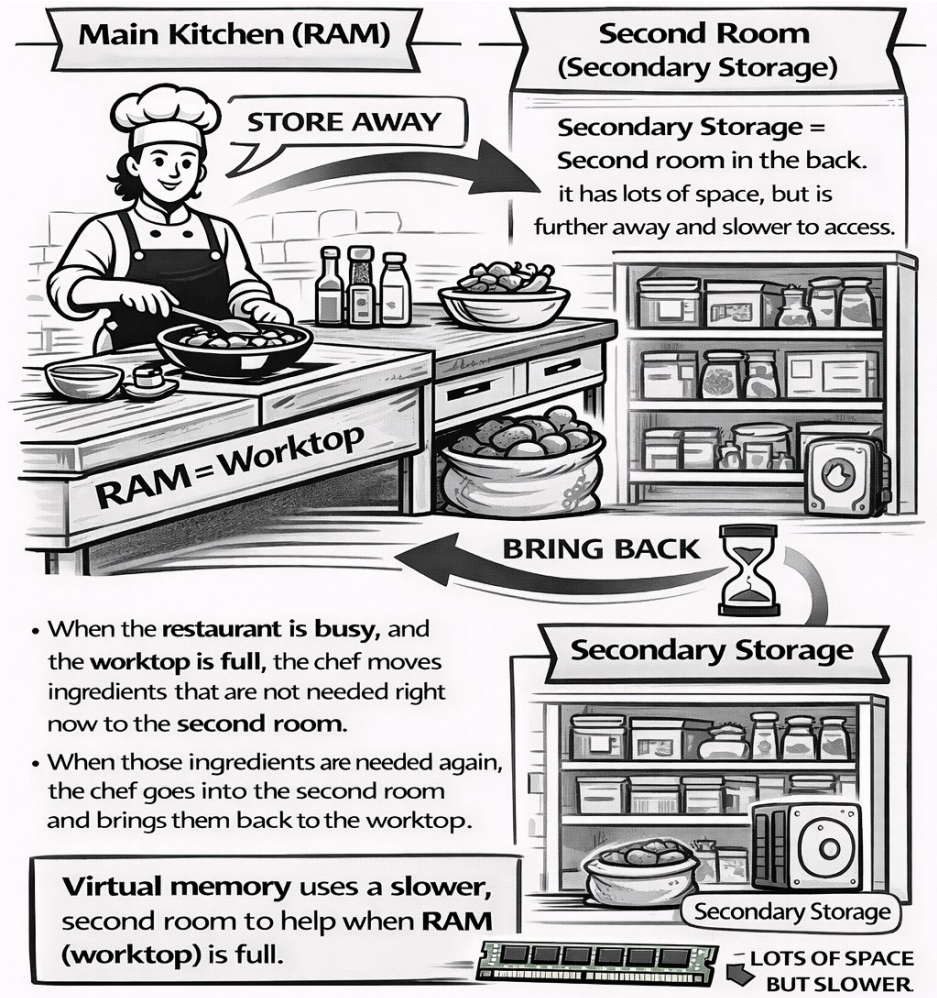
1.2.1 Primary storage (memory)	
☐ The need for primary storage ☐ The difference between RAM and ROM ☐ The purpose of ROM in a computer system ☐ The purpose of RAM in a computer system ☐ Virtual memory	**Required** ✓ Why computers have primary storage (memory) ▪ How this usually consists of RAM and ROM ✓ Key characteristics of RAM and ROM ✓ Why virtual memory may be needed in a system ✓ How virtual memory works ▪ Transfer of data between RAM and HDD when RAM is full

Computers have primary storage (memory) so they can store data and instructions that the CPU needs to access quickly while the computer is running apps.
RAM is volatile memory (means forget data without power) used to store data and programs currently in use, while ROM is non-volatile memory (means never forgets data) used to store start-up instructions.



Characteristic	RAM (Random Access Memory)	ROM (Read-Only Memory)
Purpose	Stores programs and data currently being used	Stores start-up instructions
Volatile	✔️ Data is lost when power is turned off	✘ Data is kept when power is off
Read / Write	Read and write	Read only
Speed	Very fast access	Slower than RAM
Size	Usually large (GB's)	Usually small (MB's)
What it stores	Running apps, open files, instructions in use	Start-up instructions to get the computer ready

Virtual memory is used when the computer runs out of primary storage - RAM.
If there is not enough RAM to hold all the apps and data being used, the computer moves some data out of RAM to a slower storage area (secondary storage) so the system can keep working This makes the computer act like it has more RAM than it really does but slow it down.



Teacher Tip – Common misconception

Cache improves performance by storing frequently used data close to the CPU, while virtual memory allows a system to continue running when RAM is full by using secondary storage.

If a student ever writes:

"Virtual memory is a type of cache"

❌ That is **wrong** and will lose marks.

1.2.1 Exam Questions on Memory

M	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is RAM?	RAM is the main memory that stores programs and data currently being used.	Include **currently being used**.
1	Define	What does *volatile* mean?	Volatile means data is lost when the power is switched off.	Volatile = **forgets without power**.
1	Define	What is ROM used for?	ROM stores the start-up instructions used when the computer boots.	Mention **start-up / boot**.
2	Explain	State two differences between RAM and ROM.	RAM is volatile and can be read/write, whereas ROM is non-volatile and read-only.	Two differences = two marks (e.g. volatile + read/write).
2	Explain	State two things stored in RAM.	Running programs and data currently being used, such as open files.	Give two examples: **apps** + **open files**.
3	Explain	Explain why computers need RAM.	RAM stores data and instructions currently in use so the CPU can access them quickly while programs are running.	Use the phrase **quick access for the CPU**.
3	Explain	Explain why ROM is non-volatile and why that matters.	ROM is non-volatile so it keeps its contents when power is off, meaning the computer can still load the start-up instructions when turned on.	Link **non-volatile** to **boot ing**.
3	Explain	Explain what virtual memory is.	Virtual memory is when the computer uses part of secondary storage to store data temporarily when RAM is full.	Must say **uses secondary storage when RAM is full**.
4	Describe roles	Describe two differences between RAM and ROM.	RAM stores programs and data currently being used and is volatile. ROM stores start-up instructions and is non-volatile.	Two differences + full sentences = 4 marks.
4	Describe roles	Describe what happens when a computer uses virtual memory.	When RAM is full, the system moves some data from RAM to secondary storage to free up RAM. This allows more programs to run but is slower.	Mention **move data**, **secondary storage**, **slower**.
6	Discuss / Explain	Explain the purpose of virtual memory and the effect it has on performance.	Virtual memory is used when there is not enough RAM to hold all running programs and data. The system moves some data from RAM to secondary storage so the computer can continue working. This makes it behave like it has more RAM, but performance becomes slower because secondary storage is much slower than RAM.	For 6 marks: **purpose + process + effect (slower)**.

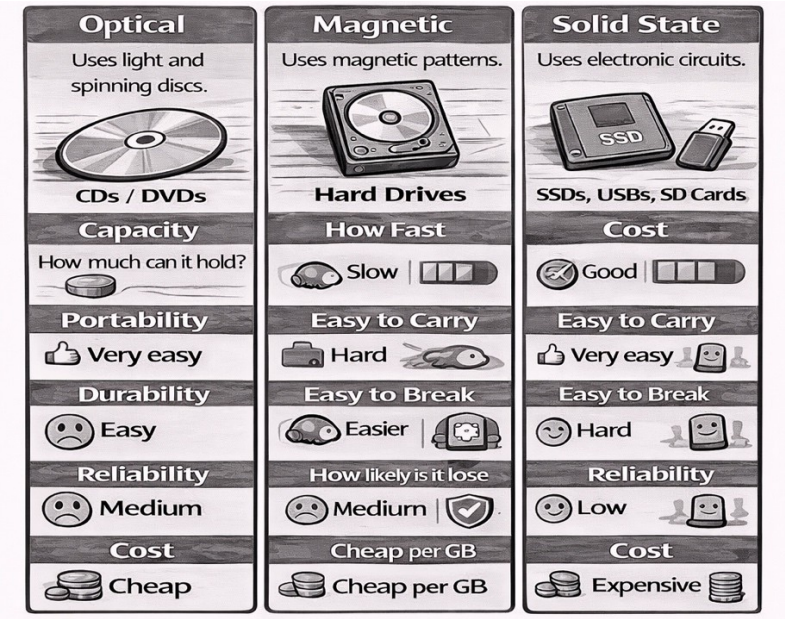
1.2.2 Secondary Storage

1.2.2 Secondary storage	
☐ The need for secondary storage ☐ Common types of storage: ○ Optical ○ Magnetic ○ Solid state ☐ Suitable storage devices and storage media for a given application ☐ The advantages and disadvantages of different storage devices and storage media relating to these characteristics: ○ Capacity ○ Speed ○ Portability ○ Durability ○ Reliability ○ Cost	**Required** ✓ Why computers have secondary storage ✓ Recognise a range of secondary storage devices/media ✓ Differences between each type of storage device/medium ✓ Compare advantages/disadvantages for each storage device ✓ Be able to apply their knowledge in context within scenarios **Not required** × Understanding of the component parts of these types of storage

Primary storage (RAM and ROM) helps the computer work right now, but it cannot store everything forever. This is why computers need secondary storage.

Recap
RAM is the computer's working space → It is fast but forgets everything when the computer is turned off.
ROM helps the computer start up → It does not store your work or files.

Secondary storage is needed to keep work, files, photos, games, and programs safe even when the computer is turned off (non-volatile).There are three main types of secondary storage, and all of them store data using binary (0s and 1s).Optical storage uses light to read data from a **spinning** disc, magnetic storage uses magnetic patterns on a **spinning** disc, and solid-state storage uses electronic circuits and has **no moving parts**.



Storage Type	Capacity	Speed	Portability	Durability	Reliability	Cost
Optical	Low	Slow	High	Low (scratches)	Medium	Cheaper than Magnetic and Solid State
Magnetic	Very high	Medium	Low	Low (moving parts)	Medium	Cheaper than SSD , more expensive the Optical
Solid State	Medium–High	Very fast	Very high	High (no moving parts)	High	More expensive optical and magnetic

1.2.2 Exam Questions on Secondary Storage

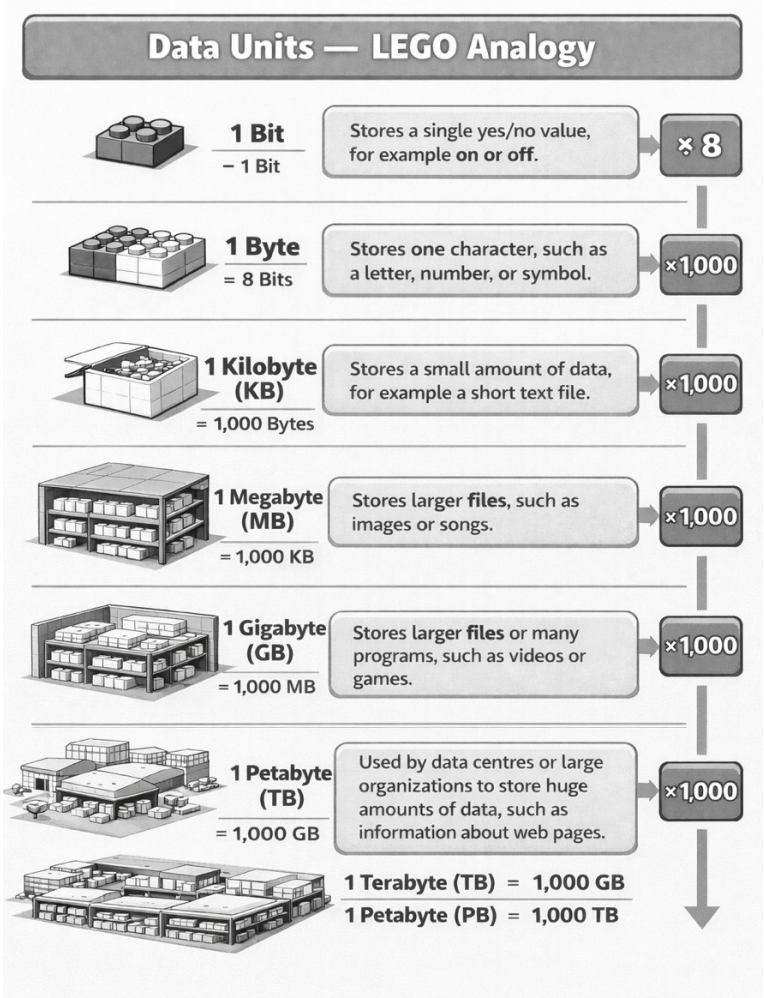
M	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is secondary storage?	Secondary storage is non-volatile storage used to store data and programs permanently.	Use the word **non-volatile**.
1	Define	Give one example of secondary storage.	Hard disk drive / SSD / USB flash drive / CD / DVD.	Just name one device.
1	Define	What does non-volatile mean?	Non-volatile means data is kept when the power is switched off.	Non-volatile = **keeps data without power**.
2	Explain	State two reasons computers need secondary storage.	It stores files permanently and keeps data when the computer is turned off.	Two clear reasons = two marks.
2	Explain	State two differences between primary and secondary storage.	Secondary storage is non-volatile and slower, while primary storage is volatile and faster.	Pick two clear contrasts.
3	Explain	Explain how optical storage stores data.	Optical storage uses laser light to read data stored as pits and lands on a spinning disc.	Mention **laser** and **spinning disc**.
3	Explain	Explain how magnetic storage stores data.	Magnetic storage stores data using magnetic patterns on a spinning disc that can be read and written by a magnetic head.	Say **magnetic patterns** + **disc**.
3	Explain	Explain how solid-state storage stores data.	Solid-state storage stores data using electronic circuits and has no moving parts.	Use **no moving parts**.
4	Describe roles	Describe two advantages of solid-state storage.	Solid-state storage is very fast and durable because it has no moving parts.	Link advantage to **reason why**.
4	Describe roles	Describe two disadvantages of optical storage.	Optical storage has a low capacity and is slow to access data.	Capacity + speed = easy marks.
6	Discuss / Explain	Compare optical, magnetic and solid-state storage.	Optical storage has low capacity and slow speed but is cheap and portable. Magnetic storage has very high capacity and medium speed but has moving parts. Solid-state storage is very fast, durable and reliable but more expensive.	Structure by **storage type**. One sentence each.

1.2.3 Units

1.2.3 Units	
☐ The units of data storage: ○ Bit ○ Nibble (4 bits) ○ Byte (8 bits) ○ Kilobyte (1,000 bytes or 1 KB) ○ Megabyte (1,000 KB) ○ Gigabyte (1,000 MB) ○ Terabyte (1,000 GB) ○ Petabyte (1,000 TB) ☐ How data needs to be converted into a binary format to be processed by a computer ☐ Data capacity and calculation of data capacity requirements	**Required** ✓ Why data must be stored in binary format ✓ Familiarity with data units and moving between each ✓ Data storage devices have different fixed capacities ✓ Calculate required storage capacity for a given set of files ✓ Calculate file sizes of sound, images and text files ▪ sound file size = sample rate x duration (s) x bit depth ▪ image file size = colour depth x image height (px) x image width (px) ▪ text file size = bits per character x number of characters **Alternatives** • Use of 1,024 for conversions and calculations would be acceptable • Allowance for metadata in calculations may be used

Data is stored in binary because computers are electronic devices made from switches that can only be on or off. Binary uses 1s (on) and 0s (off) to represent these states, making data storage simple.

Data storage is built from tiny pieces, binary digits (0 or 1) that group together to make bigger units Data is measured in units that increase in size: a bit is the smallest unit, 4 bits make a nibble, 8 bits make a byte, and larger units are formed by grouping 1,000 of the previous unit (KB, MB, GB, TB, and PB).\
Let's drop the chief chef and look at LEGO!



Teacher Tip – Common misconception

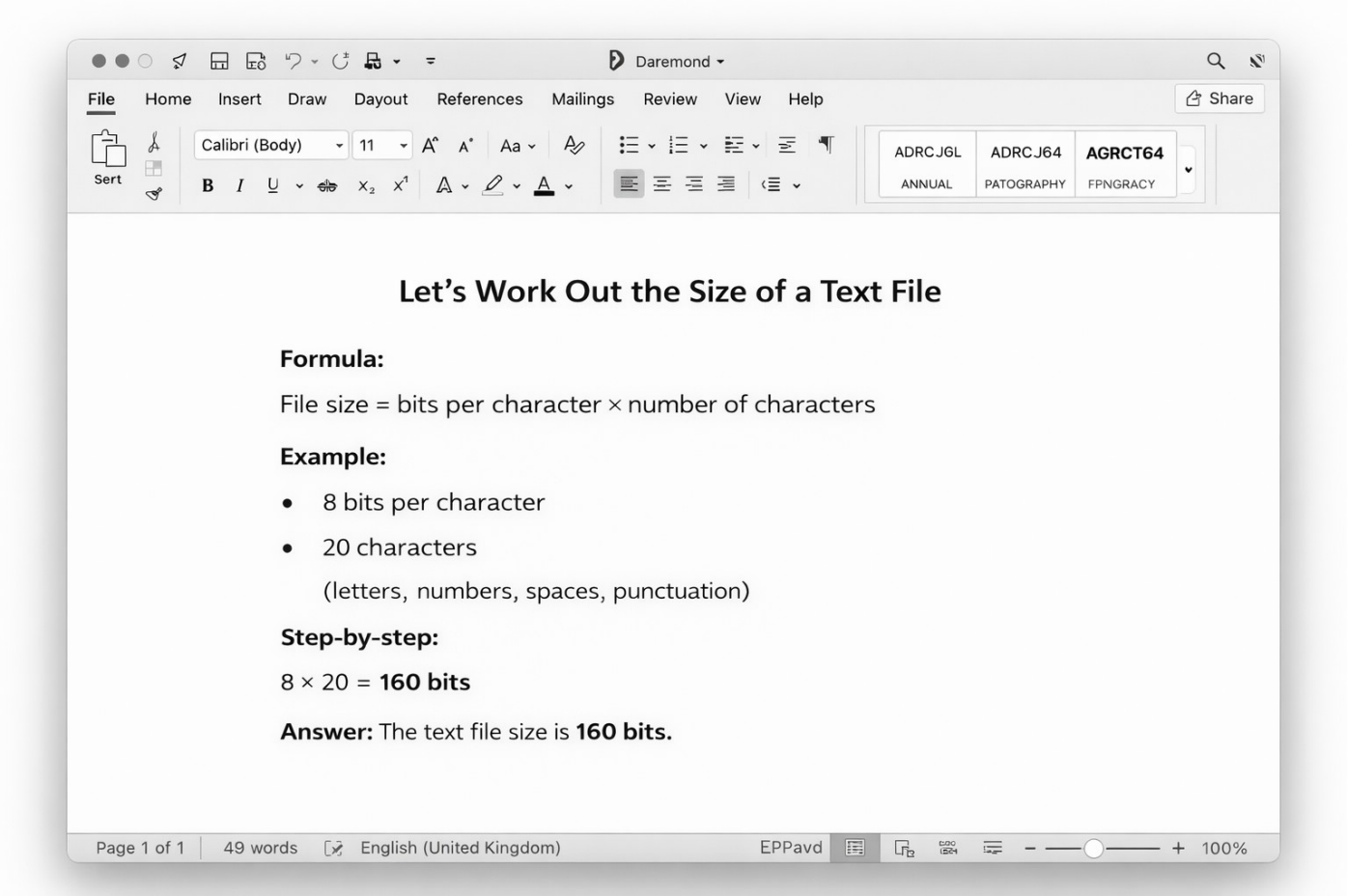
- **Capacity** = the **maximum possible storage size never changes.** Your optical, magnetic and SSD never capacity changes.
- **Free space** = how much room is left **right now** (changes as files are added or removed)

"If I empty a USB stick, does it become a bigger USB stick?"
→ **No — the capacity is fixed.**

❌ That is **wrong** and will lose marks.

Text file size depends on how much data is stored, such as the number of characters.
Let us work out the size of a text file using the formula

Text file= bits per character * number of characters

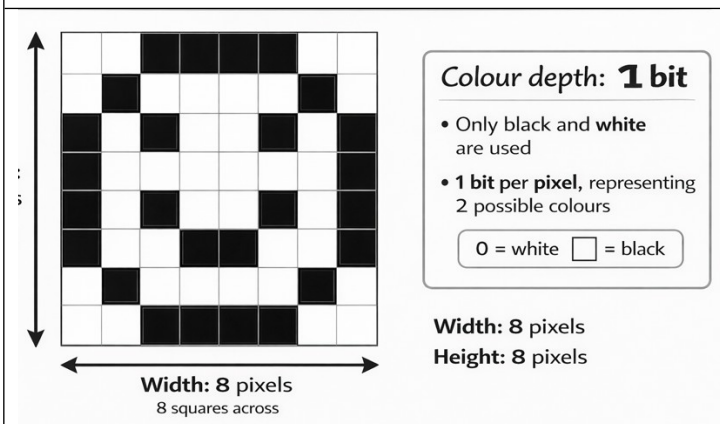
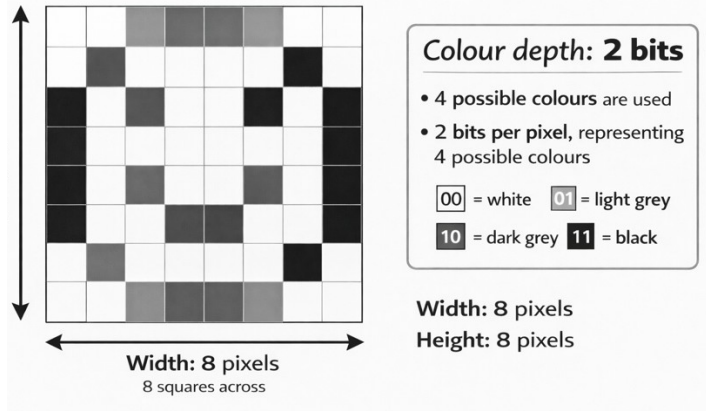
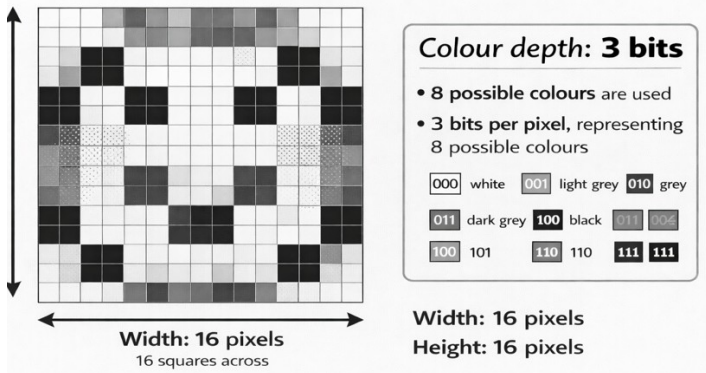


Note that the bits per character could be ASCII (8 bits) or Unicode (32-64) – Question will specify.A character set tells the computer which number matches each letter or symbol.
There are different character sets, such as ASCII and Unicode, which use different numbers of bits and support different numbers of characters.

Feature	ASCII	Unicode
Bits per character	Usually **8 bits**	Often **16 or 32 bits**
Characters supported	Basic **English letters, numbers, and symbols**	**Many languages, symbols, and emojis**
File size	**Smaller** text files	**Larger** text files
Used for	Simple English text	International text and emojis

Image file size depends on how much data is stored, such as the number of pixels and colour depth. Let us work out the size of an image file using the formula

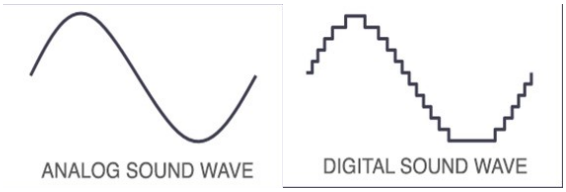
image file size in bits= colour depth (cd) x* image height (h) * image width (w)

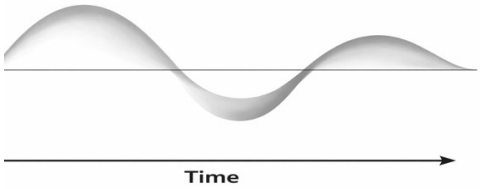
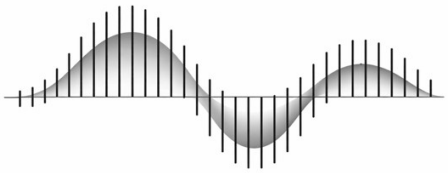
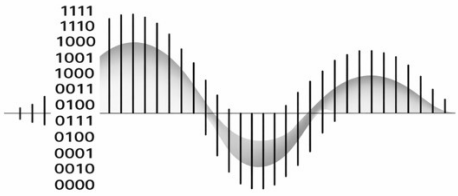
Image file	Explanation
	image file size in bits= cd * h * w = size in bits 1 * 8 * 8 = **64 bits** Or in bytes 64/8 == **8 bytes**
	image file size in bits= cd * h * w = size in bits 2 * 8 * 8 = **128 bits** Or in bytes 128/8 == **16 bytes**
	image file size in bits= cd * h * w = size in bits 3 * 16 * 16 = 768 **bits** Or in bytes 768/8 == 96 **bytes**
1000 × 800 image at **8-bit colour depth** 1) Work out pixels 1000 × 800 = **800,000 pixels** 2) Multiply by colour depth (bits per pixel) 800,000 × 8 = **6,400,000 bits** 3) Convert bits → bytes 6,400,000 ÷ 8 = **800,000 bytes** 4) Convert bytes → kilobytes 800,000 ÷ 1000 = **800 kilobytes** 5) Convert kilobytes → megabytes 800 ÷ 1000 = **0.8 megabyte**	**5000 × 2000** image at **8-bit colour depth** 1) Work out pixels 5000 × 2000 = **10,000,000 pixels** 2) Multiply by colour depth 10,000,000 × 8 = **80,000,000 bits** 3) Convert bits → bytes 80,000,000 ÷ 8 = **10,000,000 bytes** 4) Convert bytes → kilobytes 10,000,000 ÷ 1000 = **10,000 kilobytes** 5) Convert kilobytes → megabytes

	10,000 ÷ 1000 = **10 megabytes**

Analogue sound is smooth and continuous, with no steps, and is how sound exists in the real world.

Digital sound is how computers store sound by taking many snapshots (samples) and saving them as binary (0s and 1s) because computers cannot store sound/analogue waves.



Term	Definition	Image
Duration	How long the sound lasts (time along the bottom) Measured in **seconds**	
Sample rate	How often the sound is measured each second (how close the vertical sample lines are) Measured in **Hertz**	
Bit depth	How detailed each measurement is (how tall the steps can be) The number of 1's and 0's used to represent height, like colour depth! Measured in **1's and 0's**.	

Let us work out the size of a sound file

Sound file size depends on how much data is stored, such as the sample rate, duration and bit depth.

Let us work out the size of a sound file using the formula

sound file size = sample rate * duration (s) * bit depth

A sound clip has: **Sample rate:** 10 samples per second **Duration:** 5 seconds **Bit depth:** 8 bits Step-by-step	A sound clip has: **Sample rate:** 20 samples per second **Duration:** 100 seconds **Bit depth:** 8 bits Step-by-step 20 × 100 × 8 = 16,000 bits	Higher **sample rate** → bigger file Longer **duration** → bigger file Higher **bit depth** →

10 × 5 × 8 = 400 bits 400 bits /8 = 50 bytes	16,000 / 8 = 2,000 bytes 2000/1000 = 2 kilobytes	bigger file

1.2.3 Exam Questions on Units

M	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is a bit?	A bit is a binary digit that is either 0 or 1.	1 mark = short definition.
1	Define	How many bits are in a byte?	8 bits.	Memorise: **8 bits = 1 byte**.
1	Define	What is meant by capacity?	Capacity is the maximum amount of data a storage device can hold.	Capacity is **fixed**(doesn't change).
2	Explain	State two differences between capacity and free space.	Capacity is the maximum storage size and stays the same, while free space is how much room is left and changes as files are added or removed.	Two contrasts = two marks (**fixed vs changes**).
2	Explain	State two differences between ASCII and Unicode.	Unicode supports many more characters (including different languages and emojis) and usually uses more bits per character than ASCII.	Think: **more characters**+ **more bits**.
3	Explain	Explain why computers store data in binary.	Computers use electronic switches that are either on or off. Binary represents these states using 1 and 0, making it reliable for storage and processing.	Mention **on/off** and **1/0**.
3	Explain	Explain what a character set is.	A character set is a mapping that tells the computer which binary number represents each character, such as letters or symbols.	Use the word **mapping** or **code**.
3	Explain	Explain how to calculate the size of a text file.	Text file size = number of characters × bits per character, then convert bits into bytes and larger units if needed.	Always write the **formula**first.
4	Describe / Calculate	Calculate the size of a text file with 200 characters using ASCII.	ASCII is 8 bits per character. 200 × 8 = 1600 bits. 1600 ÷ 8 = 200 bytes.	Step marks: **multiply then ÷8**.
4	Describe / Calculate	Calculate the size of an image that is 1000×800 at 8-bit colour depth (in MB).	Pixels = 1000×800 = 800,000. Bits = 800,000×8 = 6,400,000. Bytes = ÷8 = 800,000. KB = ÷1000 = 800 KB. MB = ÷1000 = 0.8 MB.	Write conversion chain: **bits → bytes → KB → MB**.
6	Discuss / Explain	Explain how sample rate, bit depth and duration affect the size of a sound file.	Sound file size depends on sample rate, duration and bit depth. A higher sample rate takes more measurements per second, so the file is larger. A longer duration stores more seconds of sound, so the file is larger. A higher bit depth stores more detail per sample, so the file is larger.	For 6 marks: define each factor + link to **bigger file**.

1.2.4 Data storage

1.2.4 Data storage

Numbers

- ☐ How to convert positive denary whole numbers to binary numbers (up to and including 8 bits) and vice versa
- ☐ How to add two binary integers together (up to and including 8 bits) and explain overflow errors which may occur
- ☐ How to convert positive denary whole numbers into 2-digit hexadecimal numbers and vice versa
- ☐ How to convert binary integers to their hexadecimal equivalents and vice versa
- ☐ Binary shifts

Required

- ✓ Denary number range 0 – 255
- ✓ Hexadecimal range 00 – FF
- ✓ Binary number range 00000000 – 11111111
- ✓ Understanding of the terms 'most significant bit', and 'least significant bit'
- ✓ Conversion of any number in these ranges to another number base
- ✓ Ability to deal with binary numbers containing between 1 and 8 bits
  - e.g. 11010 is the same as 00011010
- ✓ Understand the effect of a binary shift (both left or right) on a number
- ✓ Carry out a binary shift (both left and right)

Let us convert 5,13,255 and 254 into binary. We will need the **number line. 128-64-32-16-8-4-2-1!**

Denary (Base 10)	8-bit Binary (Base 2)	How it works
5	128 64 32 16 8 4 2 1 0 0 0 0 0 1 0 1	To convert denary to binary, start with the number on left and ask "do I need this to make the number 5?" If yes write 1, if no write 0; for the number 5 we skip 128, 64, 32, 16, and 8, take 4, skip 2 because it would make 6, and take 1, giving 4 + 1 = 5.
13	128 64 32 16 8 4 2 1 0 0 0 0 1 1 0 1	13 we skip 128, 64, 32, and 16 because they are too big, take 8, take 4 to make 12, skip 2 because it would make 14, and take 1, giving 8 + 4 + 1 = 13. 8 + 4 + 1 = **13**
255	128 64 32 16 8 4 2 1 1 1 1 1 1 1 1 1	We check each place value from **left** (Most significant bit – MSB) to right and only use the numbers we need to make the total. 128+ 64+ 32 +16+ 8 +4 + 2 + 1 = **255**
254	128 64 32 16 8 4 2 1 1 1 1 1 1 1 1 0	Sometimes it better to work back from 255 if it's a number over 200! 128+ 64+ 32 +16+ 8 +4 + 2 + 0 = **254**

Let us convert 10000001, 1101, 101111 and 1111110 into decimal. We will need the **number line. 128-64-32-16-8-4-2-1!**

Binary (Base 2)	Denary (Base 10)	How it works
10000001	128 64 32 16 8 4 2 1 1 1	Put in the numbers to the number line and add them up. Simple. **128+1 = 129**
1101	8 4 2 1 1 1 1	Sometimes there will not be 8 bits. Create the number line to match the binary digits. So, for binary number line would be 8 4 2 1 **8 + 4 + 1 = 13**

101111	<table><tr><td>32</td><td>16</td><td>8</td><td>4</td><td>2</td><td>1</td></tr><tr><td>1</td><td></td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>	The **32 + 8 +4 + 2 + 1 = 47**
1111110	<table><tr><td>128</td><td>64</td><td>32</td><td>16</td><td>8</td><td>4</td><td>2</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td></tr></table>	128+ 64+ 32 +16+ 8 +4 + 2 + 0 = **254**Sometimes it better to work back from 255 if it's a number over 200!

What is hexadecimal (hex)?
Hexadecimal is a number system that uses 16 symbols instead of 10.
The 16 hex symbols Hex uses:
0–9 (just like denary) 0 1 2 3 4 5 6 7 8 9
A–F for values bigger than 9. 10 is **A**, 11 is **B**, 12 is **C**, 13 is **D**, 14 is E and 15 is **F**.

Hexadecimal is easier for humans to read than binary because it uses fewer symbols, so long binary numbers become much shorter. This is especially useful in programs like Photoshop, where colours are written using hex codes. Example: Green in Photoshop Green colour in hex: `#00FF00` What this means:`00` → no red, `FF` → lots of green and `00` → no blue. Way easier to type of say over the phone to your designer.

Common misconception
❌ *"Hex replaces binary"*
✅ **No — it just makes binary easier to read. Computers are digital, 1's and 0's only.**

Let us convert some numbers from denary (base 10) to hexadecimal (base 16). We will have to visit binary on the way. We need the **number line. 128-64-32-16-8-4-2-1 again!**

Denary (Base 10)	8-bit Binary (Base 2)	How it works - 5 Steps
254	<table><tr><td>**128**</td><td>**64**</td><td>**32**</td><td>**16**</td><td>**8**</td><td>**4**</td><td>**2**</td><td>**1**</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>**8**</td><td>**4**</td><td>**2**</td><td>**1**</td><td>**8**</td><td>**4**</td><td>**2**</td><td>**1**</td></tr><tr><td colspan="4">8+4+2+1 =15</td><td colspan="4">8+4+2=14</td></tr><tr><td colspan="4">**F**</td><td colspan="4">**E**</td></tr></table>	Step 1 – Number line Step 2 – Work out binary for **254** - 1111 1110 Step 3– Two Simple number lines 8-4-2-1 Step 4 – Calculate LHS /Calculate RHS LHS is 8+4+2+1 = 15. RHS is 8+4+2 = 14. Step 5 – Look up HEX values for LHS 15 is **F** and RHS 14 is E.
163	<table><tr><td>**128**</td><td>**64**</td><td>**32**</td><td>**16**</td><td>**8**</td><td>**4**</td><td>**2**</td><td>**1**</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>**8**</td><td>**4**</td><td>**2**</td><td>**1**</td><td>**8**</td><td>**4**</td><td>**2**</td><td>**1**</td></tr><tr><td colspan="4">**8+2 =10**</td><td colspan="4">2+1=3</td></tr><tr><td colspan="4">**A**</td><td colspan="4">**3**</td></tr></table>	Step 1 – Number line Step 2 – Work out binary for **163** - 1010 0011 Step 3– Two Simple number lines 8-4-2-1 Step 4 – Calculate LHS /Calculate RHS LHS is 8+2 = 10. RHS is 2+1=3 Step 5 – Look up HEX values for LHS 10 is **A** and RHS **3 ==3.**
103	<table><tr><td>**128**</td><td>**64**</td><td>**32**</td><td>**16**</td><td>**8**</td><td>**4**</td><td>**2**</td><td>**1**</td></tr></table>	Step 1 – Number line

	0 1 1 0 0 1 1 1 **8 4 2 1 8 4 2 1** **4+2 =6** 4+2+1 **6** **7**	Step 2 – Work out binary for **163** 1010 0011 Step 3– Two Simple number lines 8-4-2-1 Step 4 – Calculate LHS /Calculate RHS LHS is 4+2 = 6. RHS is 4+2+1 =7 Step 5 – Look up HEX values for LHS 6==6 RHS 7 ==7.

Let us convert some numbers from hexadecimal (base 16) to denary (base 10). We will have to visit binary (base 2) on the way! We will need the **number line again. 128-64-32-16-8-4-2-1!**

Denary (Base 10)	8-bit Binary (Base 2)	How it works - 5 Steps
A4	A==10 4==4 8 4 2 1 8 4 2 1 1 0 1 0 0 1 0 0 128 64 32 16 8 4 2 1 1 0 1 0 0 1 0 0 128+32+4 =164	Step 1 – Look up HEX values. For LHS A is 10 and RHS 4 is 4. Step 2 – Write the LHS/RHS number line 8-4-2-1 Step 3– Place bits, LHS / RHS. Step 4 – Write the full number line 128-64-32-16-8-4-2-1 Step 5 – Add them up A4 is 164 A4 in HEX ==164 Denary
70	7==7 0==0 8 4 2 1 8 4 2 1 0 1 1 1 0 0 0 0 128 64 32 16 8 4 2 1 0 1 1 1 0 0 0 0 64+32+16 =112	Step 1 – Look up HEX values. For LHS 7 is 7 and RHS 0 is 0. Step 2 – Write the LHS/RHS number line 8-4-2-1 Step 3– Place bits, LHS / RHS. Step 4 – Write the full number line 128-64-32-16-8-4-2-1 Step 5 – Add them up A4 is 112 70 in HEX ==112 Denary
FA	F==15 A==10 8 4 2 1 8 4 2 1 1 1 1 1 1 0 1 0 128 64 32 16 8 4 2 1 1 1 1 1 1 0 1 0 128+64+32+16+8+2 =250	Step 1 – Look up HEX values. For LHS 7 is 7 and RHS 0 is 0. Step 2 – Write the LHS/RHS number line 8-4-2-1 Step 3– Place bits, LHS / RHS. Step 4 – Write the full number line 128-64-32-16-8-4-2-1 Step 5 – Add them up FA is 250 FA in HEX ==250 Denary

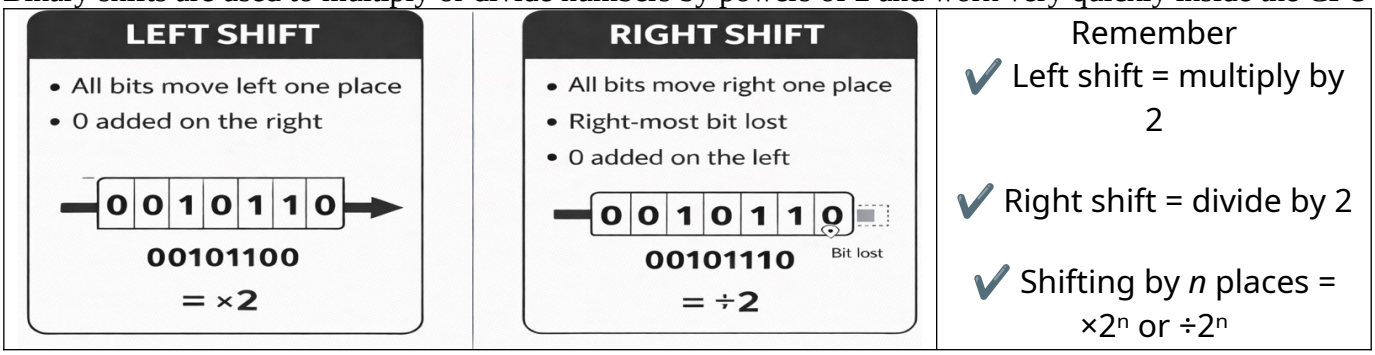
Key takeout – Use this gris in the exam – Remember A -10 B-11 C-12 D-13 E-14 F-15

128	**64**	**32**	**16**	**8**	**4**	**2**	**1**	Step 1 – Number line
								Step 2 – Work out binary
8	**4**	**2**	**1**	**8**	**4**	**2**	**1**	Step 3– Two Simple number lines 8-4-2-1
								Step 4 – Calculate LHS / RHS in denary
								Step 5 – Lookup LHS / RHS in Hex table A -10 B-11 C-12 D-13 E-14 F-15

What is a binary shift? (J277 definition)

A binary shift moves all the bits in a binary number to the left or to the right.

Binary shifts are used to multiply or divide numbers by powers of 2 and work very quickly inside the CPU



Characters

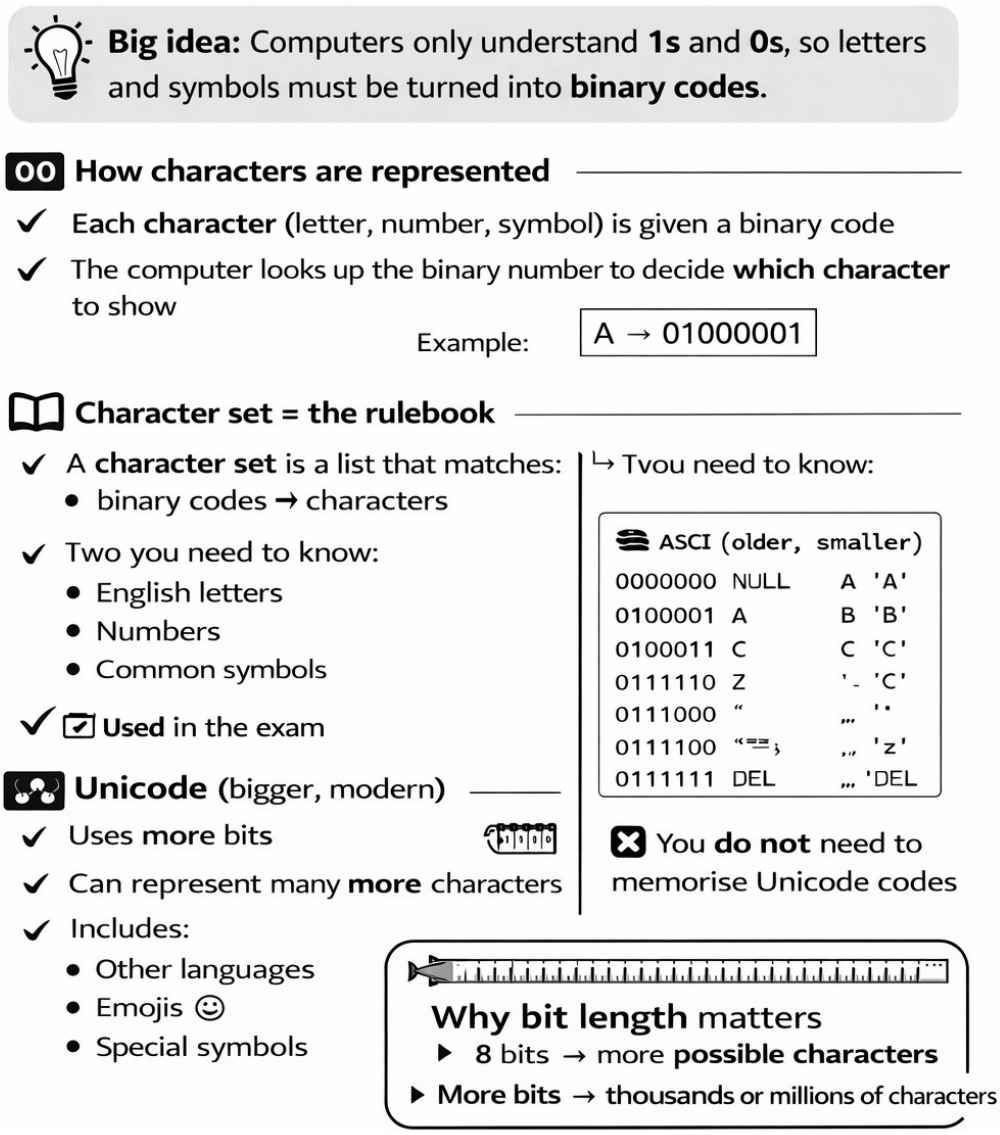
- ☐ The use of binary codes to represent characters
- ☐ The term 'character set'
- ☐ The relationship between the number of bits per character in a character set, and the number of characters which can be represented, e.g.:
  - ASCII
  - Unicode

Required

- ✓ How characters are represented in binary
- ✓ How the number of characters stored is limited by the bits available
- ✓ The differences between and impact of each character set
- ✓ Understand how character sets are logically ordered, e.g. the character code for 'B' will be one more than the character code for 'A'
- ✓ Binary representation of ASCII in the exam will use 8 bits

Not required

- ✖ Memorisation of character set codes



The ASCII code for the letter **C** is: 01000011
Using this information, work out the ASCII code for the word: DEF

Step 1: Understand the sequence

ASCII letters are stored in **alphabetical order**

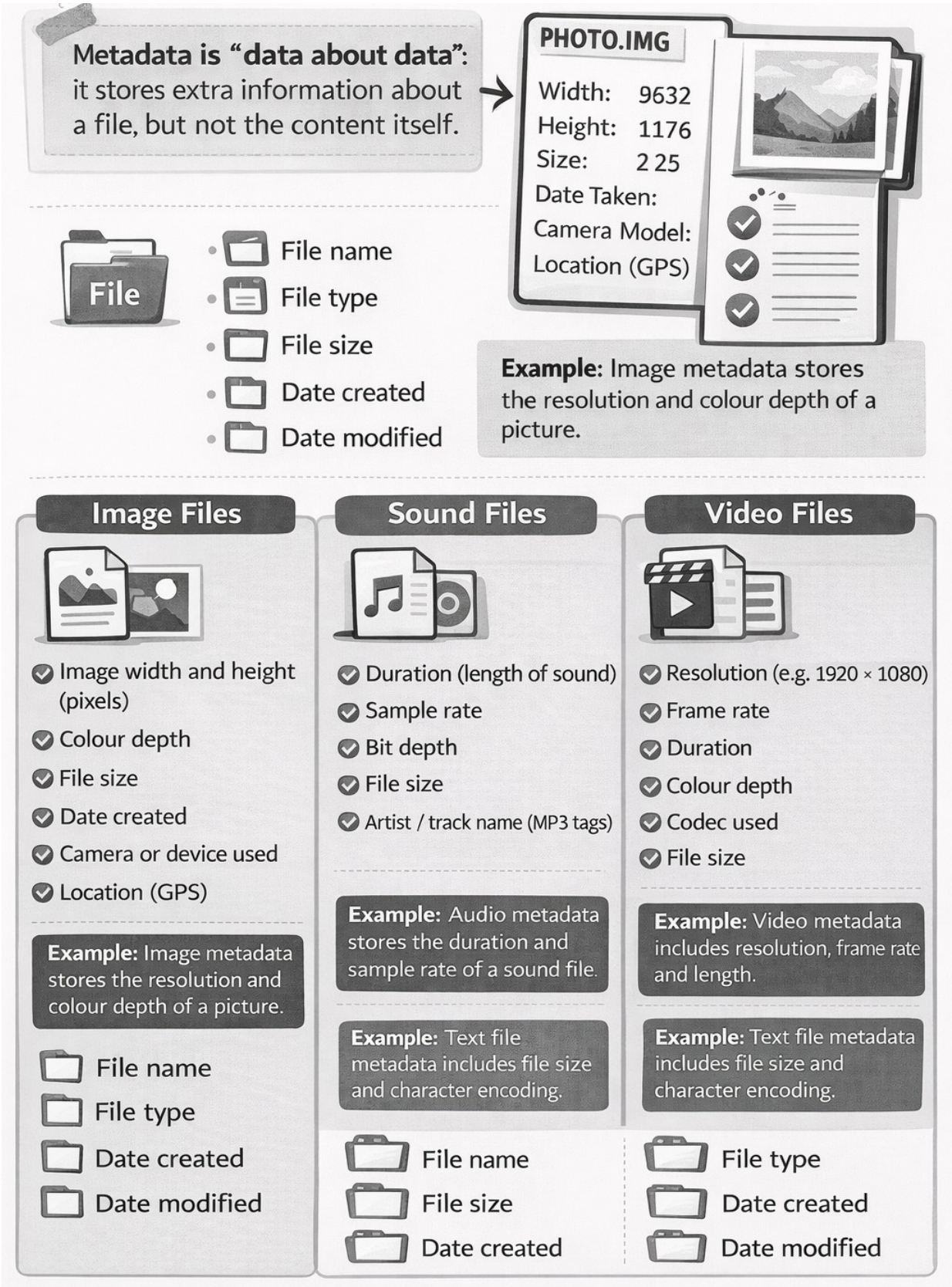
Each next letter increases the binary value by **1**

Step 2: Move forward in the sequence

C (01000011) → D (01000100) → E (01000101) → F (01000110)

Step 3: Write the answer D = 01000100 E = 01000101 F = 01000110

Metadata – Useful for web searches. The spiders talk to them!



Exam application – mixed number skills: Convert 01101101 to denary and two-digit hexadecimal. Then show an 8-bit two-place left shift of 00010110. State how the shift changes its denary value, assuming no significant bits are lost.

1.2.4 Exam questions on Data storage

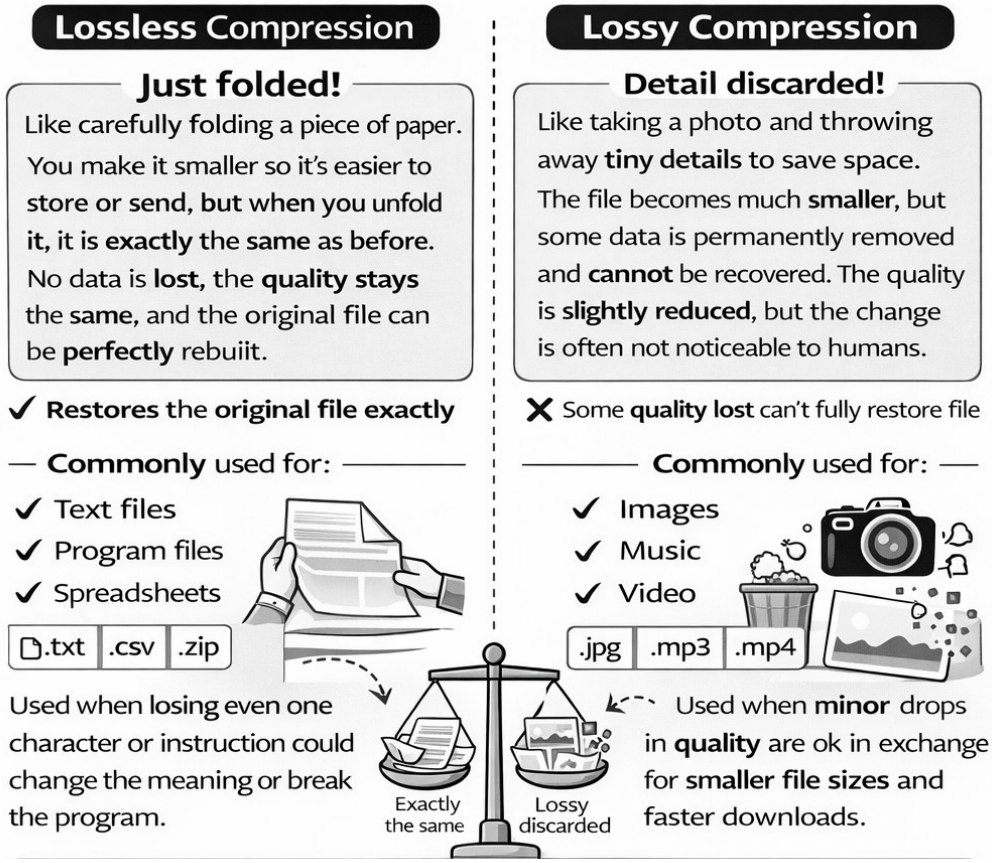
M	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is binary?	Binary is a base-2 number system that uses only 0s and 1s.	Mention **base 2** or **0 and 1**.
1	Define	What does	Letters A–F to represent values 10–15.	Memorise **A=10**

		hexadecimal use instead of only numbers?		**… F=15**.
1	Define	What is a binary shift?	A binary shift moves all the bits in a binary number left or right.	Short definition only.
2	Explain	State two reasons hexadecimal is used.	Hexadecimal shortens long binary numbers and is easier for humans to read.	Think **shorter**+ **r eadable**.
2	Explain	State two rules of binary shifts.	A left shift multiplies by 2 and a right shift divides by 2.	Left = ×2, Right = ÷2.
3	Explain	Explain how to convert a denary number into binary.	Start with the highest place value and decide if each value is needed, writing 1 if used and 0 if not.	Use the **number line** method.
3	Explain	Explain how to convert binary to denary.	Add together the place values where there is a 1 in the binary number.	Only add columns with a **1**.
3	Explain	Explain the effect of a left binary shift by 2 places.	A left shift by 2 places multiplies the number by 4 (2^2).	Shifting by n = $\times 2^n$.
3	Explain	Explain what happens to bits that move off the end in a binary shift.	Bits that move off the end are lost and cannot be recovered.	Use the word **lost**.
3	Explain	Explain what metadata is and give one example of metadata stored in a file.	Metadata is data about a file rather than the content itself. For example, image metadata can store the width and height of a picture so it can be displayed correctly.	Define metadata **and** giv e one clear example. Two parts = three marks.
3	Explain	Explain why hexadecimal does not replace binary.	Computers still use binary internally, but hexadecimal makes binary easier for humans to read.	Hex = **human-friendly**.
4	Describe / Calculate	Convert 13 into 8-bit binary.	13 = 00001101 because 8 + 4 + 1 = 13.	Show working using **8-4-2-1**.
4	Describe / Calculate	Convert the hexadecimal number A4 into denary.	A = 10 and 4 = 4. Binary is 10100100. 128 + 32 + 4 = 164.	Always lookup hex values first.
4	Describe / Apply	The ASCII code for C is 01000011. Work out the ASCII codes for D, E and F.	D = 01000100, E = 01000101, F = 01000110.	ASCII letters increase by **1 in binary**.
6	Discuss / Explain	Explain how denary numbers are converted into hexadecimal.	Convert denary to 8-bit binary, split into two 4-bit groups, convert each group using 8-4-2-1, then replace values 10–15 with A–F.	Structure: **binary → split → lookup**.

1.2.5 Compression

1.2.5 Compression	
☐ The need for compression ☐ Types of compression: ○ Lossy ○ Lossless	**Required** ✓ Common scenarios where compression may be needed ✓ Advantages and disadvantages of each type of compression ✓ Effects on the file for each type of compression **Not required** × Ability to carry out specific compression algorithms

Compression reduces file size so data takes up less storage and transfers faster.



Feature	Lossless	Lossy
Data lost?	❌ No	✅ Yes
File restored exactly?	✅ Yes	❌ No
File size reduction	Smaller	Much smaller
Used for	Text, programs	Images, sound, video

1.2.5 Exam Questions on Compression

M	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is compression?	Compression reduces the size of a file so it takes up less storage space.	Use the phrase **reduces file size**.
1	Define	What is lossless compression?	Lossless compression reduces file size without losing any data.	Say **no data lost**.
1	Define	What is lossy compression?	Lossy compression reduces file size by permanently removing some data.	Say **data is lost**.
2	Explain	State two benefits of compression.	Compression reduces storage space needed and allows files to be transferred faster.	Storage + speed = easy marks.
2	Explain	State two differences between lossless and lossy compression.	Lossless compression does not lose data and can be restored exactly, while lossy compression loses data and cannot be restored exactly.	Two contrasts = two marks.
3	Explain	Explain why lossless compression is used for text files.	Text files must be restored exactly, as losing data could change words or instructions.	Mention **exact restoration**.
3	Explain	Explain why lossy compression is suitable for images, sound, or video.	Some data can be removed without noticeable loss of quality, greatly reducing file size.	Say **quality loss not noticeable**.
3	Explain	Explain how compression improves data transfer.	Smaller files require less data to be sent, so they transfer more quickly over a network.	Link **smaller file**to **faster transfer**.
4	Describe roles	Describe the difference between lossless and lossy compression.	Lossless compression reduces file size without losing data and allows exact restoration, while lossy compression permanently removes some data and cannot be restored exactly.	Define both clearly.
4	Describe roles	Describe two situations where lossy compression would be used.	Lossy compression is used for images and video where a small reduction in quality is acceptable.	Give two examples.
6	Discuss / Explain	Explain the advantages and disadvantages of lossy compression.	Lossy compression greatly reduces file size, improving storage and transfer speeds. However, data is permanently lost and quality may be reduced, meaning the original file cannot be restored exactly.	For 6 marks: **pros + cons**.

1.3 Computer networks, connections and protocols

Keyword	Definition
Network	Two or more computers connected for the purposes of sharing data or resources
Local Area Network	A network which covers a small geographic area such as a single building.
Wide Area Network	A network which covers a large geographic area.
The internet	A global network connecting millions of computers and other devices.
Bandwidth	The amount of data which can be transferred via a network at any given time.
Network Topology	The way in which computers and other network devices are connected.
Encryption	Encryption is the process of converting plain text (readable data) into cipher text (scrambled, unreadable data) using an algorithm and a key, so that only authorised users can access it.
HTTP/HTTPS	A protocol used to transfer web pages (HTTPS is secure and uses encryption)
SMTP	A protocol used to send emails
POP	A protocol used to receive emails and download them to one device (old school!)
IMAP	A protocol used to receive emails and store them on a server so they can be accessed on multiple devices (new school – we all have so many devices now)
TCP	Puts data into packets and ensures they are sent and received correctly
IP	Identifies devices using IP addresses and ensures data is sent to the correct destination
Star	A network where all devices connect to a central device (e.g. switch) Cheap, like a school network!
Mesh	A network where devices are connected to many or all other devices. Expensive, like Apple/Google/Microsoft/Amazon HQ!
Cloud Services	Storing and accessing data and programs over the internet instead of on a local device

1.3.1 Networks and Topologies

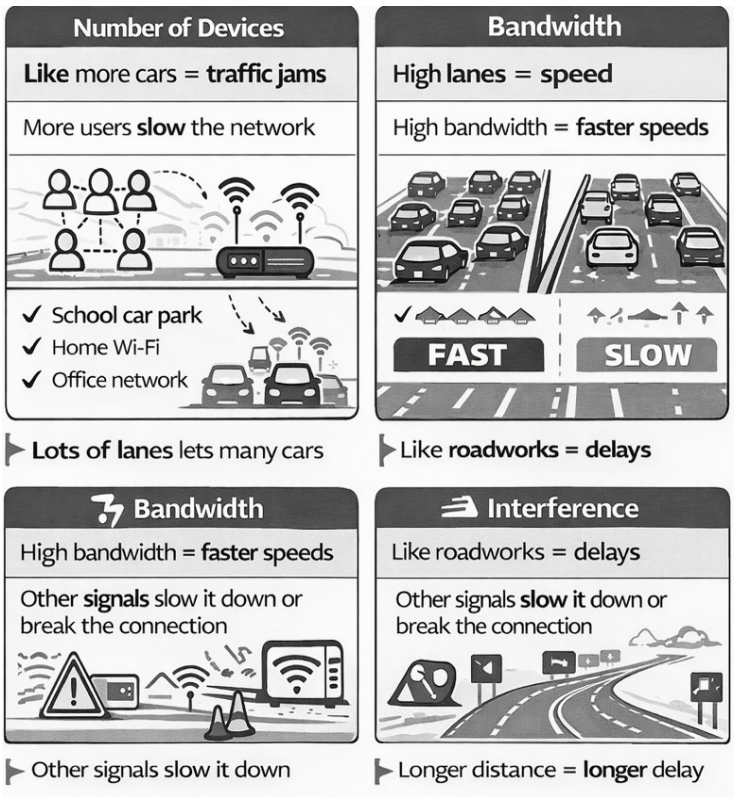
1.3.1 Networks and topologies	
☐ Types of network: ○ LAN (Local Area Network) ○ WAN (Wide Area Network) ☐ Factors that affect the performance of networks ☐ The different roles of computers in a client-server and a peer-to-peer network ☐ The hardware needed to connect stand-alone computers into a Local Area Network: ○ Wireless access points ○ Routers ○ Switches ○ NIC (Network Interface Controller/Card) ○ Transmission media ☐ The Internet as a worldwide collection of computer networks: ○ DNS (Domain Name Server) ○ Hosting ○ The Cloud ○ Web servers and clients ☐ Star and Mesh network topologies	**Required** ✓ The characteristics of LANs and WANs including common examples of each ✓ Understanding of different factors that can affect the performance of a network, e.g.: ▪ Number of devices connected ▪ Bandwidth ✓ The tasks performed by each piece of hardware ✓ The concept of the Internet as a network of computer networks ✓ A Domain Name Service (DNS) is made up of multiple Domain Name Servers ✓ A DNS's role in the conversion of a URL to an IP address ✓ Concept of servers providing services (e.g. Web server → Web pages, File server → file storage/retrieval) ✓ Concept of clients requesting/using services from a server ✓ The Cloud: remote service provision (e.g. storage, software, processing) ✓ Advantages and disadvantages of the Cloud ✓ Advantages and disadvantages of the Star and Mesh topologies ✓ Apply understanding of networks to a given scenario

What is a network?

A network can be defined as "two or more computers connected together to share information and resources". This can involve physical or wireless connections, or both.

Network concept	Road analogy	What it means
LAN (Local Area Network)	Small car park or private road	Covers a short distance, has few users, and traffic moves quickly
Example of LAN	School car park	A school network is small, local, and fast because everyone is close together
WAN (Wide Area Network)	Motorway system	Covers long distances, connects many networks, and is slower and more expensive
Example of WAN	Motorways between cities	The Internet connects many places across the world

Network Performance



The different roles of computers in a client-server and a peer-to-peer network

Client–Server Computers have **different roles**: a **server** provides services (files, printers, logins) and **clients** request and use them

Peer-to-Peer All computers have the **same role** and can both **share and receive** files

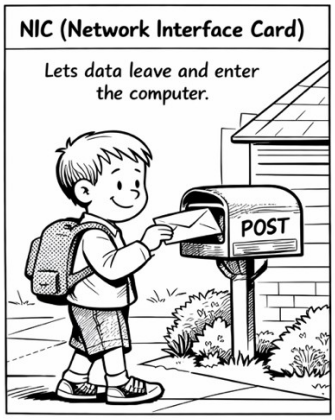
Advantages

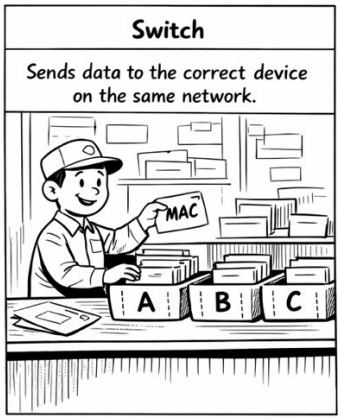
Client–Server	Peer-to-Peer
Central control and management	Easy and cheap to set up
Stronger security and permissions	No need for a dedicated server
Data backed up in one place	Works well for very small networks
Files available even if clients are off	Simple file sharing between computers

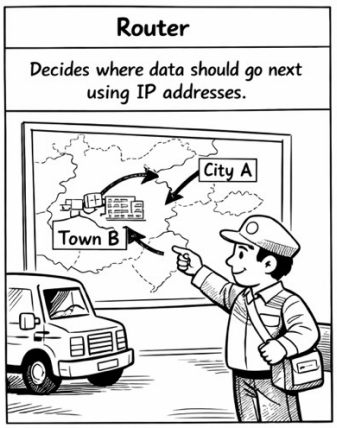
Disadvantages

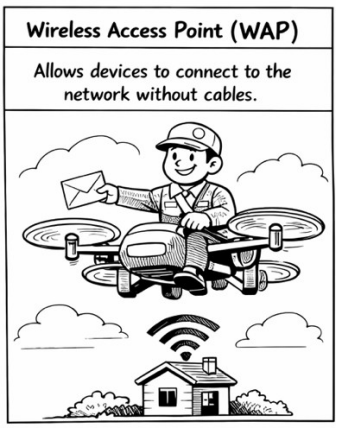
Client–Server	Peer-to-Peer
More expensive (server hardware/software)	Weaker security
Needs specialist setup and maintenance	No central backup
Server failure affects whole network	Files unavailable if a computer is off
More complex to manage	Poor performance on larger network

The hardware needed to connect stand-alone computers into a Local Area Network









1 **NIC** (Network Interface Card) — the starting point. Every device on a network has a NIC.
A NIC allows a computer to connect to a network and send and receive data.
Each **NIC** has a **MAC** address.
A MAC address uniquely identifies a device on a network. 12 hexadecimal characters
Example: -**A4:5E:60:9F:12:3B**
Assigned by the manufacturer/Used inside a LAN

2 **Switch** - delivering data **inside the LAN**
When data leaves the NIC, it goes to a switch.
A switch connects devices within a LAN and sends data to the correct device using MAC addresses.
What the switch does: Reads the MAC address
Sends data only to the intended device
Improves performance (not broadcast to everyone)

3 **Router** - moving between networks(LANS)
If the data needs to leave the local network, it goes to a router. A router forwards data between different networks using IP addresses.
Key role: Uses IP addresses, not MAC addresses
Connects a LAN to the Internet
Decides the best route

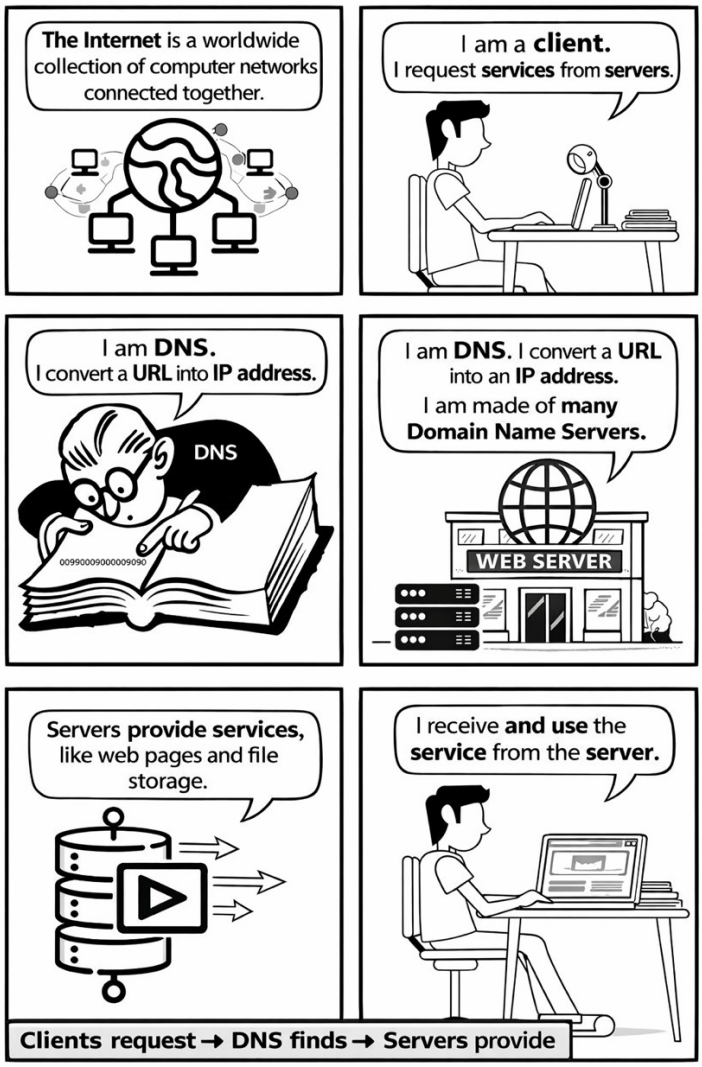
4 Wireless Access Point (WAP)
A WAP allows wireless devices to join the network.
A wireless access point allows devices to connect to a network without cables.
Important**: A WAP** does not replace **a switch or router**. It connects into the wired network.
Wireless devices still have: A NIC /A MAC address /An IP address

Data leaves a device through the NIC, **is sent to the correct device by a switch using MAC addresses**, routed **between networks by a router using IP addresses**, and can be delivered wirelessly using a wireless access point

Transmission Media

Type	Uses	Advantages	Disadvantages
Copper cable	Electrical signals	Cheap, reliable	Slower than fibre
Fibre optic	Light signals	Very fast, long distance	Expensive
Wireless	Radio waves	No cables, flexible	Interference, security

The Internet as a worldwide collection of computer networks:



Exam-perfect

The Internet is a network of computer networks where clients request services from servers, DNS converts URLs into IP addresses, and servers provide services such as web pages and file storage.

DNS Converts URLs – web address www.apple.com

To numbers(**IPv4**) or Hexadecimal (**IPv6**)

18.132.191.179 (IPv4) – old format **IPv4**
2001:0db8:85a3:0000:0000:8a2e:0370:7334 - – new format **IPv6**

A **client** is a device that **requests** and uses services provided by a server. Examples of clients include: Laptops / Smartphones/ Tablets /Desktop computers. Clients do not store the main shared data. Instead, they send requests, such as asking for a web page or a file

A **server** is a computer that **provides** services to clients over a network. Servers are usually: **Always** switched **on** / More **powerful** than client computers / Designed to **handle many requests** at the same time

Different servers provide types of services:

Server type	**Service provided**
Web server	Stores and provides web pages
File server	Stores files and allows file storage and retrieval
Email server	Sends and receives emails

DNS (Domain Name Service) is a system that converts human-readable website names into IP addresses. Computers do not understand website names (URL) like: www.bbc.co.uk. They communicate using IP addresses instead.

DNS is made up of multiple servers

DNS is not a single server — it is made up of many Domain Name Servers distributed around the world. This makes a DNS: **Faster** /More **reliable** / Less likely to fail

A client enters a URL into a web browser
The request is sent to a DNS server
DNS looks up the corresponding IP address
The IP address is returned to the client
The client then contacts the web server using that IP addressDNS allows users to access **websites using names instead of numbers (IPv4) or hex (IPv6).**

The Cloud: remote service provision

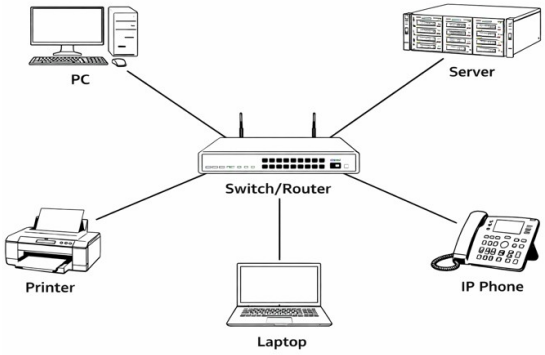
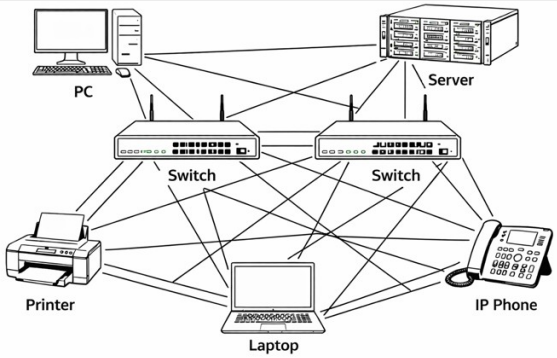
The Cloud means using the Internet to store files, run software, or do processing on computers somewhere else. When you use apps, social media, or streaming, the work is done on big computers far away, not on your own device, that's the Cloud. For example, when you do school work online, your phone or laptop is using the Cloud to do the hard work for your device! There are three categories. Storage, software and processing.

Cloud Services

Misconception	Examples of web services	Service
❌ *"Social media pics are stored on my phone"* ✅ Pics and videos **are stored on cloud servers**	Snapchat / Insta / WhatsApp – chats , images and videos	Cloud Storage
❌ *"The game runs fully on my phone"* ✅ **Your phone sends inputs; the servers do the hard work**	Clash Royale / Fortnite / AI	Cloud Processing
❌ No (can run in a browser) ✅ **Your phone sends inputs; the servers do the hard work**	YouTube / Netflix /Discord - If you can use it by opening a web browser and logging in, it is cloud software and needs no installation	Cloud Software

Star and Mesh network topologies

In a star topology used in schools such as Reepham High, all devices connect to a central switch, making the network easy to manage but dependent on the central device.The Internet uses a mesh-like topology so data can travel along multiple routes, meaning if one connection fails the network can reroute data and continue working without a single point of failure.

Star (Reepham High School)	**Mesh** (The Internet / Apple HQ)
simple	*high reliability*
affordable	*fault tolerance*
easy to manage.	*allow multiple routes for data and avoid single points of failure*
	

Exam application – IPv4 and DNS: Decide which are valid IPv4 addresses and give a reason: 10.2.14.7, 172.16.300.4 and 192.168.1.1. Then explain the sequence from entering a website name to the browser receiving its IP address. Remember: four decimal octets, each 0–255; DNS performs name resolution.

Devices (or switches/routers) are connected to multiple others Data can travel directly between nodes or via alternative routes This makes them fault tolerant. Star networks, such as those used in schools like Reepham High School, are simple, affordable, and easy to manage but rely on a central device. In contrast, mesh networks, such as the Internet or large organisations like Apple HQ, provide high reliability and fault tolerance by allowing multiple routes for data and avoiding single points of failure.

1.3.1 Exam Questions on Networks and Topologies

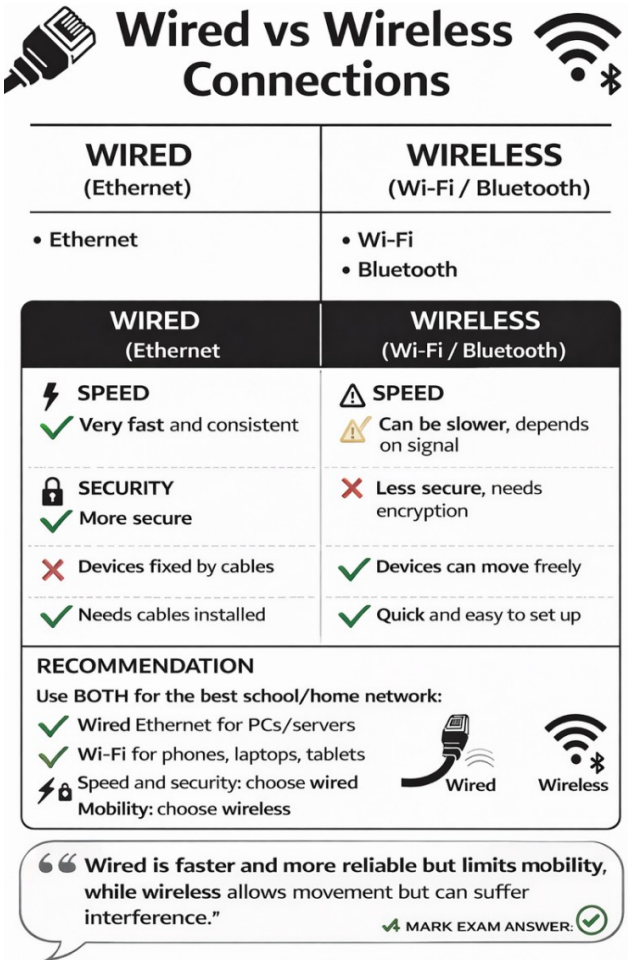
M	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is a computer network?	A network is two or more computers connected together to share data and resources.	Use **two or more** and **share**.
1	Define	What does LAN stand for?	Local Area Network.	Expand the acronym only.
1	Define	What does WAN stand for?	Wide Area Network.	No explanation needed.
2	Explain	State two differences between a LAN and a WAN.	A LAN covers a small area and is fast, while a WAN covers a large area and is slower.	Pick **distance** + **speed**.
2	Explain	State two uses of a computer network.	Networks allow file sharing and sharing hardware such as printers.	Two uses = two marks.
3	Explain	Explain the difference between a client–server network and a peer-to-peer network.	In a client–server network, servers provide services and clients request them. In a peer-to-peer network, all computers have the same role.	Different roles vs same role.
3	Explain	Explain the role of a Network Interface Card (NIC).	A NIC allows a device to connect to a network and send and receive data, and it has a unique MAC address.	Say **connect**, **send/receive**, **MAC**.
3	Explain	Explain the role of a switch in a LAN.	A switch connects devices in a LAN and sends data only to the intended device using MAC addresses.	Switch = **MAC addresses**.
3	Explain	Explain the difference between copper cable and fibre-optic cable.	Copper uses electrical signals and is slower, while fibre uses light signals and is much faster over long distances.	Fibre = **light + fast**.
3	Explain	Explain one disadvantage of wireless transmission.	Wireless transmission can suffer from interference and security issues.	Mention **interference**or **security**.
3	Explain	Explain what	DNS converts human-readable	DNS = **name → IP**.

		DNS is used for.	website names into IP addresses so computers can locate servers.	
3	Explain	Explain what is meant by the Cloud.	The Cloud is the use of remote servers on the Internet to store data, run software, or process data.	Say **remote servers**.
3	Explain	Explain one advantage of a star topology.	Star networks are easy to manage because all devices connect to a central switch.	Star = **central point**.
3	Explain	Explain one advantage of a mesh topology.	Mesh networks are fault tolerant because data can travel along multiple routes.	Mesh = **multiple routes**.
4	Describe roles	Describe the role of a router in a network.	A router forwards data between different networks using IP addresses and connects a LAN to the Internet.	Router = **between networks + IP**.
4	Describe roles	Describe two advantages of a client–server network.	Client–server networks allow central management and stronger security, with data backed up in one place.	Two advantages, one sentence each.
4	Describe roles	Describe two disadvantages of a peer-to-peer network.	Peer-to-peer networks have weaker security and no central backup, and files may be unavailable if a computer is off.	Security + availability.
4	Describe	Describe how DNS works.	A client sends a URL to a DNS server, which looks up the IP address and returns it so the client can contact the web server.	Write steps in order.
4	Describe	Describe two cloud services.	Cloud storage stores files online, while cloud software allows applications to run in a web browser without installation.	Storage vs software.
6	Discuss / Explain	Explain how the Internet works as a client–server network.	The Internet is a network of networks where clients request services from servers. DNS converts URLs into IP addresses, and servers provide services such as web pages and file storage.	Use **client**, **server**, **DNS**, **services**.
6	Discuss / Explain	Compare star and mesh network topologies.	Star networks are simple and easy to manage but rely on a central device. Mesh networks allow multiple routes, making them fault tolerant and reliable but more complex and expensive.	One paragraph per topology.

1.3.2 Wired and Wireless Networks

1.3.2 Wired and wireless networks, protocols and layers	
☐ Modes of connection: ○ Wired • Ethernet ○ Wireless • Wi-Fi • Bluetooth ☐ Encryption ☐ IP addressing and MAC addressing ☐ Standards ☐ Common protocols including: ○ TCP/IP (Transmission Control Protocol/Internet Protocol) ○ HTTP (Hyper Text Transfer Protocol) ○ HTTPS (Hyper Text Transfer Protocol Secure) ○ FTP (File Transfer Protocol) ○ POP (Post Office Protocol) ○ IMAP (Internet Message Access Protocol) ○ SMTP (Simple Mail Transfer Protocol) ☐ The concept of layers	**Required** ✓ Compare benefits and drawbacks of wired versus wireless connection ✓ Recommend one or more connections for a given scenario ✓ The principle of encryption to secure data across network connections ✓ IP addressing and the format of an IP address (IPv4 and IPv6) ✓ A MAC address is assigned to devices; its use within a network ✓ The principle of a standard to provide rules for areas of computing ✓ Standards allows hardware/software to interact across different manufacturers/producers ✓ The principle of a (communication) protocol as a set of rules for transferring data ✓ That different types of protocols are used for different purposes ✓ The basic principles of each protocol i.e. its purpose and key features ✓ How layers are used in protocols, and the benefits of using layers; for a teaching example, please refer to the 4-layer TCP/IP model **Not required** × Understand how Ethernet, Wi-Fi and Bluetooth protocols work × Understand differences between static and dynamic, or public and private IP addresses × Knowledge of individual standards × Knowledge of the names and function of each TCP/IP layer

Modes of Connection



Encryption

Encryption is the process of **scrambling data into an encoded form** so that only authorised users can understand it. When data is sent across a **network connection** (especially the internet or Wi-Fi), it could be intercepted. Encryption ensures that **even if data is intercepted, it cannot be read** without the correct key.

Connection Type	Why encryption is needed
Wi-Fi	Data travels through the air and is easy to intercept
Internet	Data passes through many routers
Bluetooth	Short range but still vulnerable

Standards and the TCP/IP Layers & Protocols

Standards provide **agreed rules** that allow hardware and software from different manufacturers to work together. Communication protocols are sets of rules for transferring data across a network. **Different protocols** are used for **different purposes**, such as HTTP for viewing web pages, SMTP for sending emails, and FTP for transferring files.

Network communication is split into **layers** so that each layer has a **specific job**, making networks easier to design, manage, and fix. For example, sending a letter happens in stages. Writing it, putting it in an envelope, delivering it, and transporting it.

Layer	Protocols	What it does	Correct Letter Analogy
Application	HTTP, HTTPS, SMTP, FTP	What service is being used	Writing & reading the letter
Transport	TCP	Checks delivery & order	Making sure all pages arrive
Internet	IP	Addressing & routing	Choosing the delivery route
Link (network access)	Ethernet, Wi-Fi	Physical transmission	Letterboxes, vans, roads

TCP/IP Layers & Protocols

(Posting a Letter Analogy)

Overview: Why layers?

Network communication is split into **layers** so that each layer has a **specific** job, making networks easier to design, manage and fix.

Analogy: Sending a letter happens in stages—writing it, putting it in an envelope, delivering it, and transporting it.

Layer	Main Protocols	What it does (simple)	Letter
Application Layer	HTTP, HTTPS, FTP, SMTP, POP, IMAP	How applications use the network	**Writing &** reading
Transport TCP	TCP	Ensures **reliable** delivery	**Making sure** all pages arrive
Internet Layer	IP	Handles **addressing & routing**	**Choosing the** delivery route
Network Access	Ethernet, Wi-Fi	Sends data physically	Letterboxes, vans, roads

1 Application Layer

- **HTTP, HTTPS,** FTP, SMTP, POP, IMAP
- How applications use the network

2 Transport Layer

- **TCP**
- Ensures **reliable** delivery

3 Internet Layer

- IP
- Handles **addressing** & routing

4 Network Access Layer

- **Ethernet,** Wi-Fi
- Sends data physically

Quick memory trick: A T I N

- **Application** — what is being sent
- **Transport** — delivery checked
- **Internet** — plan the route
- **Network access** — how it travels

> **The TCP/IP** model uses layers, where application protocols such as HTTP and SMTP define how data is used, TCP ensures reliable delivery, IP handles addressing and routing, and the network access layer transnits data physically.

Protocol	Purpose	Letter Analogy
HTTP	View websites	Reading a letter
HTTPS	Secure websites	Letter in a locked envelope
FTP	Transfer files	Sending a parcel
SMTP	Send emails	Posting a letter
POP	Download emails	Taking letters home
IMAP	Sync emails	Letters stay at the post office
TCP	Reliable delivery of data	Making sure **all pages arrive** and in the correct order
IP	Addressing & routing	Choosing the **delivery route** through sorting offices

SMTP sends emails, POP downloads emails to one device and then deletes them from the server (which can be inconvenient), while IMAP synchronises emails across multiple devices, making it the modern approach.

Note: Knowledge of the names and function of each TCP/IP layer NOT required. Included for clarity

1.3.2 Exam Questions on Wired and Wireless Networks

M	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is encryption?	Encryption is the process of scrambling data so that only authorised users can read it.	Use **scramble** and **authorised**.
1	Define	Name one communication protocol.	HTTP / HTTPS / FTP / SMTP / TCP / IP.	Just name one.
1	Define	What does HTTPS provide that HTTP does not?	Encryption / security.	HTTPS = **secure**.
2	Explain	State two reasons encryption is needed on Wi-Fi networks.	Wi-Fi data travels through the air and can be intercepted, so encryption prevents unauthorised access.	Air + interception = marks.
2	Explain	State two benefits of network standards.	Standards allow devices from different manufacturers to work together and ensure reliable communication.	Think **compatibility** + **reliability**.
3	Explain	Explain why data sent over the Internet should be encrypted.	Data passes through many routers and could be intercepted, so encryption prevents unauthorised users from reading it.	Mention **routers**+ **in tercepted**.
3	Explain	Explain the purpose of communication protocols.	Protocols are sets of rules that control how data is transmitted so devices can communicate correctly.	Use **rules** + **commu nication**.
3	Explain	Explain the difference between HTTP and HTTPS.	HTTP is used to view web pages, while HTTPS encrypts the data to provide secure communication.	HTTPS = HTTP + **encryption**.
3	Explain	Explain the difference between SMTP and POP.	SMTP is used to send emails, while POP downloads emails to one device and removes them from the server.	Send vs download.
3	Explain	Explain the difference between POP and IMAP.	POP downloads emails to one device, while IMAP synchronises emails across multiple devices.	IMAP = **sync**.
4	Describe	Describe the role of TCP in data transmission.	TCP ensures data is delivered reliably and in the correct order by checking that packets arrive successfully.	Use **reliable** and **ord er**.
4	Describe	Describe the role of IP in network communication.	IP is responsible for addressing and routing data so it reaches the correct destination.	IP = **address** + **route**.
4	Describe	Describe two differences between wired and wireless networks.	Wired networks use cables and are more reliable, while wireless networks use radio waves and are more flexible but can suffer interference.	Cable vs radio + reliability.
6	Discuss / Explain	Explain how protocols and standards allow data to be transmitted successfully across networks.	Standards ensure devices from different manufacturers can communicate. Protocols such as HTTP, TCP, and IP define how data is requested, delivered reliably, and routed to the correct destination, making communication efficient and reliable.	Mention **standards** + **protocols** + **roles**.

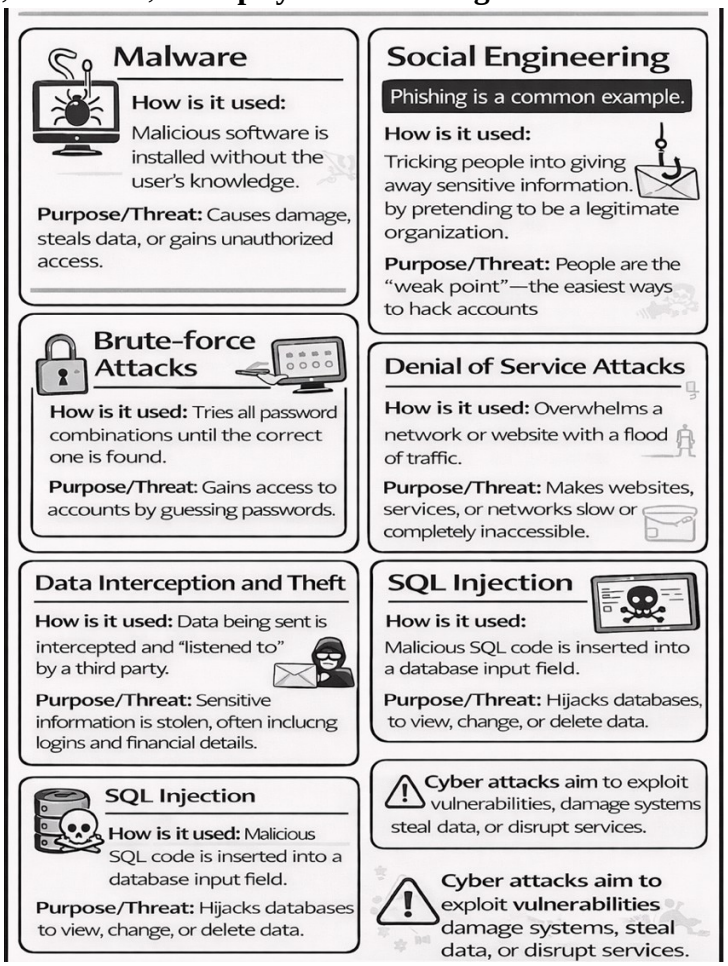
1.4 Network Security

Keyword	Definition
Brute Force Attack	A program attempting all possible password combinations.
Data Interception	Data transmission being read by an unauthorised user/program.
Malware	Software that damages/deletes data.
Anti-Malware	Detects malware on a system and alerts the user to malware if present.
Firewall	Controls incoming/outgoing traffic checking if it is authorised.
Encryption	Scrambles data using an algorithm to make it unreadable without a key.
User Access Levels	Controls users so that they can only access/edit data depending on their access level.
Password/Biometrics /Authentication	Code/Fingerprint must be correctly entered to gain access.
Two-Step authentication	A code is sent to a user's separate device to access the data this code needs to be inputted.
Social Engineering	Using deception to manipulate users to gain personal data.
Shoulder Surfing	Watching a user entering a password and using it to gain access to an account.
Phishing	Fake emails sent to a person to make the user send personal data.
Pharming	Software that directs user to a fake website where they enter their personal data.
DDOS	Too many requests are sent to a server, so it does not respond/crashes/denies access/slows access.
Hacker	Person gaining unauthorised access to a system/account.
Trojan	Malware disguised as legitimate software.
Worm	Software that replicates across a network, so it uses up all the bandwidth.
Ransomware	Encrypts/corrupts/locks access to data and data cannot be accessed until demands are met (usually money).
Physical threat	Damages to hardware that deletes/corrupts data.

1.4.1 Threats to Computer Systems & Networks

1.4.1 Threats to computer systems and networks	
☐ Forms of attack: ○ Malware ○ Social engineering, e.g. phishing, people as the 'weak point' ○ Brute-force attacks ○ Denial of service attacks ○ Data interception and theft ○ The concept of SQL injection	**Required** ✓ Threats posed to devices/systems ✓ Knowledge/principles of each form of attack including: ▪ How the attack is used ▪ The purpose of the attack

Attacks try to trick people, break passwords, steal data, or stop systems from working. Hackers either **trick people**, **break passwords**, **steal data**, or **stop systems working.**



Attack	What it is	Analogy (Easy to remember)
Malware	Harmful software that damages devices or steals data	Catching a **computer virus**
Social Engineering (Phishing)	Tricking people into giving away information	Someone **pretending to be trusted** to steal secrets
Brute-Force Attack	Trying many passwords until one works	**Trying every key** on a keyring
Denial of Service (DoS)	Overloading a system so it cannot be used	**Blocking a doorway** so no one can get in
Data Interception & Theft	Stealing data while it is being sent	**Listening in** on a private conversation
SQL Injection	Inserting malicious commands into a database	Using a **cheat code** to unlock everything

1.4.1 Exam Questions on Threats to Computer Systems & Networks

M	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is malware?	Malware is harmful software designed to damage a system or steal data.	Malware = **malicious software**.
1	Define	What is phishing?	Phishing is when users are tricked into giving away personal information.	Phishing = **trick people**.
1	Define	What is a brute-force attack?	A brute-force attack tries many passwords until the correct one is found.	Think **every password**.
2	Explain	State two examples of cyber attacks.	Malware attacks and denial of service attacks.	Just name two.
2	Explain	State two aims of a cyber attack.	To steal data and to stop systems from working.	Steal data / disrupt service.
3	Explain	Explain how a brute-force attack works.	A brute-force attack repeatedly tries many possible passwords until the correct one is found.	Mention **many attempts**.
3	Explain	Explain how phishing attacks work.	Phishing attacks trick users into clicking fake links or emails that appear genuine to steal information.	Use **fake email** or **link**.
3	Explain	Explain how a denial of service (DoS) attack affects a system.	A DoS attack overloads a system with requests so legitimate users cannot access it.	Say **overload**and **cannot access**.
3	Explain	Explain what is meant by data interception and theft.	Data interception is when data is captured while being transmitted and then stolen.	Mention **during transmission**.
4	Describe	Describe two ways malware can harm a computer system.	Malware can delete files and steal personal data such as passwords.	Two harms = two marks.
4	Describe	Describe two social engineering attacks.	Phishing emails and fake websites that trick users into giving personal information.	Give two examples.
6	Discuss / Explain	Explain why humans are often the weakest link in computer security.	Many attacks rely on tricking users through phishing or social engineering rather than technical weaknesses. If users click links or share passwords, attackers can gain access without breaking security systems.	Mention **human error**+ **social engineering**.

1.4.2 Identifying and Preventing Vulnerabilities

1.4.2 Identifying and preventing vulnerabilities	
☐ Common prevention methods: ○ Penetration testing ○ Anti-malware software ○ Firewalls ○ User access levels ○ Passwords ○ Encryption ○ Physical security	**Required** ✓ Understanding of how to limit the threats posed in 1.4.1 ✓ Understanding of methods to remove vulnerabilities ✓ Knowledge/principles of each prevention method: ▪ What each prevention method may limit/prevent ▪ How it limits the attack

Attacks can be reduced using security measures such as encryption, firewalls, strong passwords, user education, and input validation

Safety Tool	What it does (simple)	Like in real life
Anti-malware software	Finds and removes bad software	A **doctor** for your computer
User access levels	Only lets people do what they're allowed to do	Different **keys** for rooms
Strong passwords	Makes accounts hard to break into	A **strong lock**
Two-Factor Authentication (2FA)	Uses two checks to prove it's you	A lock **and** a code
User training	Teaches people how to stay safe	Learning **road safety**
Account lock-out policy	Locks accounts after too many wrong tries	A door that **locks itself**
Firewalls	Blocks bad traffic from the network	A **security guard**
Penetration testing	Finds weak spots before hackers do	A **friendly burglar**
Encryption	Scrambles messages so others can't read them	Writing in **secret code**
Input validation & sanitisation	Checks what people type to stop harm	A **safety checker**
Physical security	Protects computers in real life	**Locks, doors, CCTV**

Attack	How to stop / limit it
Malware	• **Anti-malware software** to detect and remove malicious programs • **Regular updates** to fix security weaknesses • **User access levels** to stop users installing unauthorised software
Social Engineering (Phishing)	• **User access levels** so damage is limited if an account is compromised • **Strong passwords** and **2-factor authentication** • **User awareness training** (don't click suspicious links)
Brute-Force Attack	• **Strong passwords** (long, complex, unique) • **Account lockout policies** after failed attempts • **Firewalls** to block repeated login attempts
Denial of Service (DoS)	• **Firewalls** to filter and block suspicious traffic • **Traffic monitoring** and rate limiting • **Penetration testing** to identify weaknesses before attackers do
Data Interception &	• **Encryption** (HTTPS, WPA2/WPA3) to protect data in transit • **Firewalls** to control network access • **Physical security** to prevent

Theft	unauthorised access to network hardware
SQL Injection	• **Input validation** and sanitisation • **User access levels** to limit database permissions • **Penetration testing** to find vulnerabilities in web forms

1.4.2 Exam Questions on Identifying and Preventing Vulnerabilities

M	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is a security vulnerability?	A security vulnerability is a weakness in a system that can be exploited by an attacker.	Use **weakness** + **exploited**.
1	Define	What is encryption used for?	Encryption is used to protect data by scrambling it so unauthorised users cannot read it.	Say **scramble data**.
1	Define	What is a firewall?	A firewall monitors and blocks unauthorised network traffic.	Firewall = **blocks traffic**.
2	Explain	State two ways to prevent malware attacks.	Using anti-malware software and keeping systems up to date.	Two tools = two marks.
2	Explain	State two features of strong passwords.	Strong passwords are long and contain a mix of letters, numbers, and symbols.	Length + complexity.
3	Explain	Explain how user access levels improve security.	User access levels limit what users can do, reducing damage if an account is compromised.	Use **limit access**.
3	Explain	Explain how anti-malware software protects a system.	Anti-malware software detects, quarantines, and removes malicious software.	Detect + remove.
3	Explain	Explain how account lockout policies prevent brute-force attacks.	Accounts are locked after repeated failed login attempts, stopping attackers from trying many passwords.	Mention **failed attempts**.
3	Explain	Explain how input validation helps prevent SQL injection.	Input validation checks and sanitises user input so harmful code cannot be executed.	Use **check input**.
4	Describe	Describe two ways to prevent phishing attacks.	User training helps users recognise phishing emails, and two-factor authentication limits damage if passwords are stolen.	One defence per sentence.
4	Describe	Describe two ways firewalls help protect networks.	Firewalls block unauthorised access and filter suspicious traffic entering or leaving a network.	Block + filter.
6	Discuss / Explain	Explain how a range of security measures can be used to protect a	Systems are protected using a combination of measures such as firewalls to block unauthorised access, encryption to protect data, strong passwords and account lockout	For 6 marks: **multiple measures + why**.

		computer system.	policies to prevent unauthorised logins, and user training to reduce human error. Using multiple measures reduces the risk of successful attacks.	

1.5 Software

Keyword	Definition
Operating System	Controls the operation of all aspects of the computer and provides an interface by which a user can interact with the computer.
User interface	A part of the operating system or other software which allows the user to interact with and control it.
Multitasking	The ability for an operating system to run more than one task or application at the same time.
File management	A function of the operating system which manages files and folders stored on either the computer's internal or external storage.
Defragmentation	Software which reorders files stored on a hard disk so that they run in a sensible order to improve performance and efficiency.
Data Compression	Software which compresses data so that it takes up less storage space.
Memory management	The function of the Operating System which manages the computer's memory, controlling how the memory is used and which applications can access it.
Peripheral Management	The ability of an operating system to control and manage hardware devices such as printers, mice or speakers which are connected to the computer.
Physical threat	Damages to hardware that deletes/corrupts data.

1.5.1 Operating Systems

1.5.1 Operating systems	
☐ The purpose and functionality of operating systems: ○ User interface ○ Memory management and multitasking ○ Peripheral management and drivers ○ User management ○ File management	**Required** ✓ What each function of an operating system does ✓ Features of a user interface ✓ Memory management, e.g. the transfer of data between memory, and how this allows for multitasking ✓ Understand that: ▪ Data is transferred between devices and the processor ▪ This process needs to be managed ✓ User management functions, e.g.: ▪ Allocation of an account ▪ Access rights ▪ Security, etc. ✓ File management, and the key features, e.g.: ▪ Naming ▪ Allocating to folders ▪ Moving files ▪ Saving, etc. **Not required** × Understanding of paging or segmentation

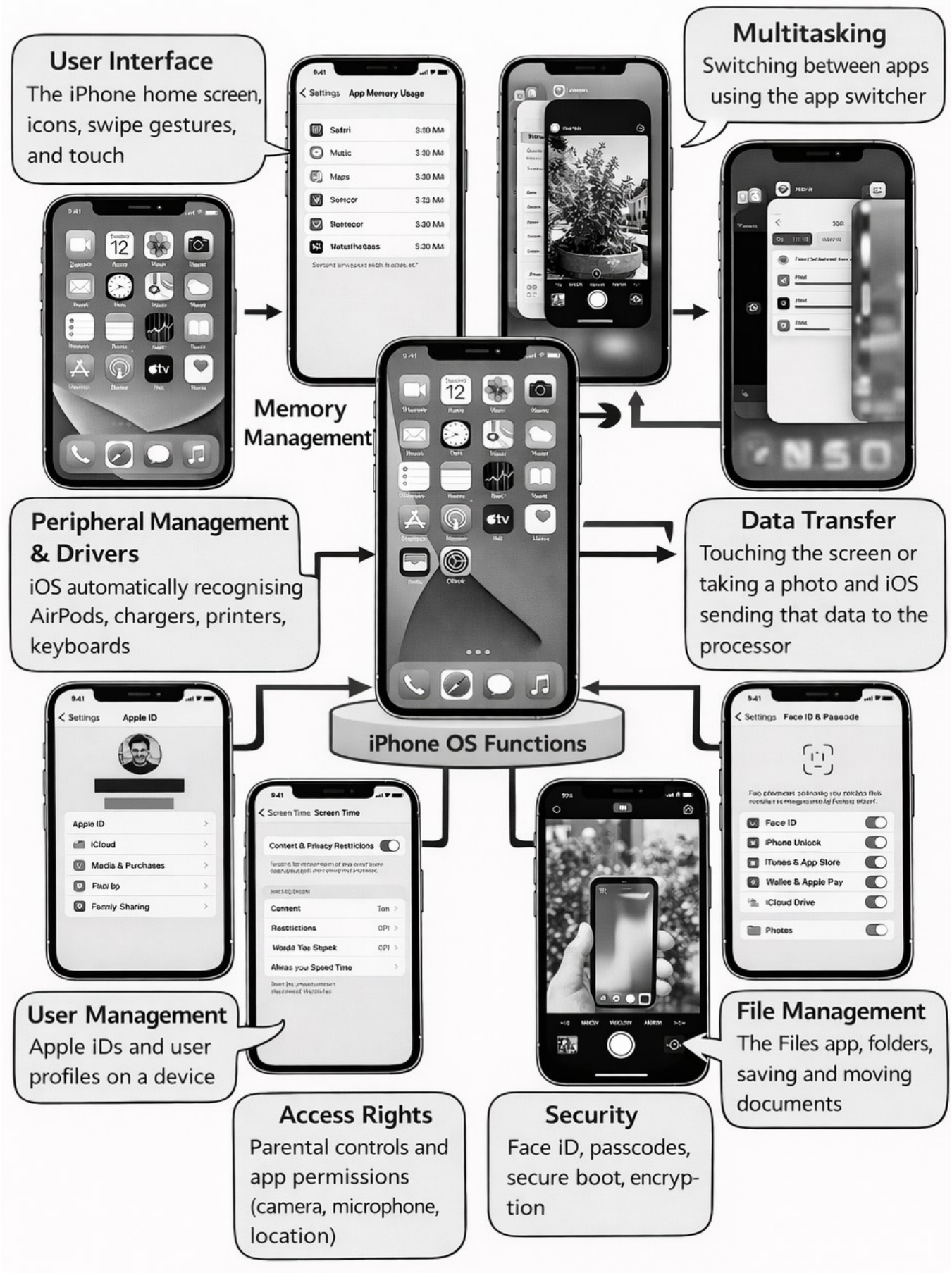
An operating system (OS) manages the computer's hardware, software, memory, files and users

OS Function	What it does	Key features / examples	Why it matters (exam focus)
User Interface (UI)	Allows the user to interact with the computer	Windows, icons, menus, pointer, touchscreen	Makes the computer usable
Memory Management	Allocates RAM to applications and transfers data between memory and the CPU	Loads programs into memory, frees memory when closed	Prevents crashes and allows programs to run
Multitasking	Allows more than one program to run at the same time	CPU switches quickly between applications	Improves efficiency and user experience
Peripheral Management & Drivers	Controls input/output devices using drivers	Keyboard, mouse, printer, display	Allows hardware to communicate with software
Data Transfer Management	Manages data sent between devices and the processor	Input and output coordination	Prevents data loss or errors
User Management	Creates and manages user accounts	Login accounts, passwords, admin vs standard users	Improves security and controls access
Access Rights	Controls what users can do	Read/write permissions, restricted settings	Protects files and system settings
Security	Protects the system and user data	Passwords, permissions, account control	Prevents unauthorised access

File Management	Organises and stores data	File naming, saving, folders, moving files	Makes files easy to find and manage

An operating system manages the user interface, memory, hardware, users and files to allow a computer to run efficiently and securely.

Let look at how you phone operating does this



OS Function	What it does	📱 iPhone Analogy
User Interface (UI)	Allows the user to interact with the computer	The **iPhone home screen**, icons, swipe gestures, and touch
Memory Management	Allocates RAM to applications and transfers data between memory and the CPU	iOS deciding **how much memory each app gets**so apps don't crash
Multitasking	Allows more than one program to run at the same time	Switching between apps using the **app switcher**
Peripheral Management & Drivers	Controls input/output devices using drivers	iOS automatically recognising **AirPods, chargers, printers, keyboards**
Data Transfer Management	Manages data sent between devices and the processor	Touching the screen or taking a photo and iOS **sending that data to the processor**
User Management	Creates and manages user accounts	**Apple IDs** and user profiles on a device
Access Rights	Controls what users can do	**Parental controls** and app permissions (camera, microphone, location)
Security	Protects the system and user data	**Face ID, passcodes, secure boot, encryption**
File Management	Organises and stores data	The **Files app**, folders, saving and moving documents

Exam application – match the OS function: Match each task to its operating-system role: creating a new user profile; allocating RAM to running programs; organising a saved folder; sending data to a printer. Choose from user, memory, file and peripheral management, and explain one match.

1.5.1 Exam Questions on Operating Systems

M	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is an operating system?	An operating system manages the computer's hardware, software, memory, files and users.	Use the word **manages**.
1	Define	Name one function of an operating system.	Memory management / File management / User interface / Security.	Just name one function.
1	Define	What is a user interface (UI)?	A user interface allows the user to interact with the computer.	UI = **how you use it**.
2	Explain	State two functions of an operating system.	The OS manages memory and controls input/output devices.	Two functions = two marks.
2	Explain	State two reasons why an operating system is important.	It allows programs to run and prevents the system from crashing.	Think **run programs** + **stability**.
3	Explain	Explain how memory management helps a computer run efficiently.	Memory management allocates RAM to applications and frees it when programs close, preventing crashes.	Mention **allocates** and **frees RAM**.
3	Explain	Explain how multitasking is handled by an operating system.	The operating system switches the CPU quickly between programs so multiple applications appear to run at the same time.	Use **CPU switching**.
3	Explain	Explain the role of device drivers in an operating system.	Device drivers allow the operating system to communicate with hardware devices such as printers and keyboards.	Drivers = **hardware communication**.
3	Explain	Explain how file management helps users.	File management allows files to be saved, organised, and retrieved easily using folders and filenames.	Mention **organise files**.
4	Describe	Describe the role of an operating system in managing users.	The OS creates and manages user accounts, controls access rights, and uses passwords to improve security.	User accounts + permissions.
4	Describe	Describe two security features provided by an operating system.	Operating systems use passwords and access rights to prevent unauthorised access to data and settings.	Two features, one sentence each.
6	Discuss / Explain	Explain the main functions of an operating system.	An operating system manages the user interface so users can interact with the computer, controls memory allocation to allow programs to run efficiently, manages hardware using drivers, organises files, and controls users and security. These functions allow the computer to run efficiently and securely.	Structure by **function**, one sentence each.

1.5.2 Utility Software

1.5.2 Utility software	
☐ The purpose and functionality of utility software ☐ Utility system software: ○ Encryption software ○ Defragmentation ○ Data compression	**Required** ✓ Understand that computers often come with utility software, and how this performs housekeeping tasks ✓ Purpose of encryption, defragmentation and data compression software and why it is required ✓ Utility software is needed to perform additional tasks that may not be carried out by an operating system

Utility software helps look after a computer by performing extra housekeeping tasks that the operating system does not always do automatically. Your phone works "out of the box", but extra tools quietly run in the background to keep it fast, safe and organised.

Utility Software	What it does (simple)	Why it is needed	📱 Mobile phone analogy
Encryption software	Scrambles data so only authorised users can read it	Protects personal and sensitive data if a device is lost, stolen or hacked	Locking your phone with **Face ID / fingerprint** so no one else can read your messages
Defragmentation	Reorganises data on a hard disk so files are stored together	Makes files load faster and improves performance on HDDs	Tidying apps and storage so your phone doesn't feel slow
Data compression	Reduces file size by removing unnecessary data	Saves storage space and speeds up file transfer	Your phone **shrinking photos or videos** to save space
Utility software (overall)	Performs extra housekeeping tasks	Keeps the system fast, secure and organised	Background tools that **quietly keep your phone running smoothly**

Utility software performs housekeeping tasks such as encryption, defragmentation and data compression to improve security, performance and storage efficiency, which are not always carried out by the operating system.

Operating System	Utility Software
Makes the device work	Looks after the device
Manages hardware, memory, users and files	Performs extra maintenance jobs
Always running	Often hidden or runs in the background

1.5.2 Exam Questions on Utility Software

M	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is utility software?	Utility software performs maintenance and housekeeping tasks to help manage and protect a computer system.	Use **maintenance** or **housekeeping**.
1	Define	Name one type of utility software.	Encryption / Defragmentation / Data compression.	Just name one.
1	Define	What is encryption software used for?	Encryption software scrambles data so unauthorised users cannot read it.	Encryption = **scramble data**.
2	Explain	State two tasks carried out by utility software.	Utility software can encrypt data and compress files.	Two tasks = two marks.
2	Explain	State two differences between an operating system and utility software.	An operating system manages hardware and users, while utility software performs maintenance tasks and often runs in the background.	OS = **runs system**, utility = **looks after system**.
3	Explain	Explain how defragmentation improves performance.	Defragmentation reorganises data on a hard disk so files are stored together, reducing access time and improving performance.	HDD only — mention **faster access**.
3	Explain	Explain why encryption software is important.	Encryption protects sensitive data so it cannot be read if a device is lost, stolen, or hacked.	Link to **lost or stolen device**.
3	Explain	Explain how data compression benefits users.	Compression reduces file size, saving storage space and allowing files to be transferred faster.	Smaller file = **faster transfer**.
4	Describe	Describe two examples of utility software.	Encryption software protects data by scrambling it, while defragmentation reorganises files to improve hard disk performance.	Two utilities, one sentence each.
4	Describe	Describe the difference between utility software and an operating	An operating system manages the core functions of a computer, while utility software performs additional maintenance tasks to improve performance and security.	Core vs maintenance.

		system.		
6	Discuss / Explain	Explain the role of utility software in maintaining a computer system.	Utility software performs housekeeping tasks such as encryption to protect data, defragmentation to improve performance on hard disks, and data compression to save storage space. These tasks help keep the system secure, efficient, and organised, and are not always carried out automatically by the operating system.	For 6 marks: **multiple utilities + why they matter**.

1.6 Impacts of Technology

Keyword	Definition
The Data Protection Act (2018)	A law in the United Kingdom which governs the processing of personal data. It outlines the requirements for organisations who handle data and the rights of those who provide the data.
Computer Misuse Act (1990)	A law in the United Kingdom which makes misuse of computer systems illegal if certain criteria are met.
Copyright Designs and Patents Act (1988)	A law in the United Kingdom which provides protection in law for works and materials. The act explains what protections are available, and what (if any) steps need to be taken to register for these protections.
Software Licenses	The right for an organisation or user to use a piece of computer software.
Ethical	The idea of right and wrong by which people should behave.
Legal	A set of rules which lay out behaviours which are not allowed and the punishments for breaking these rules.
Cultural	A set of behaviours and expectations which people accept as being normal and expected within their daily lives. These are not enforced by any laws, and don't always have to do with right and wrong.
Environmental	Matters relating to the natural world around us and making sure we take good care of things like animals, the atmosphere and the planet.
Privacy	The expectation that certain information is kept secret.
Open-Source Software	Software where the code is made available to the public to view or edit.
Proprietary Software	Software where the source code is owned by a person or organisation and is not available to the public.

1.6.1 Issues and Legislation

1.6.1 Ethical, legal, cultural and environmental impact	
☐ Impacts of digital technology on wider society including: ○ Ethical issues ○ Legal issues ○ Cultural issues ○ Environmental issues ○ Privacy issues ☐ Legislation relevant to Computer Science: ○ The Data Protection Act 2018 ○ Computer Misuse Act 1990 ○ Copyright Designs and Patents Act 1988 ○ Software licences (i.e. open source and proprietary)	**Required** ✓ Technology introduces ethical, legal, cultural, environmental and privacy issues ✓ Knowledge of a variety of examples of digital technology and how this impacts on society ✓ An ability to discuss the impact of technology based around the issues listed ✓ The purpose of each piece of legislation and the specific actions it allows or prohibits ✓ The need to license software and the purpose of a software licence ✓ Features of open source (providing access to the source code and the ability to change the software) ✓ Features of proprietary (no access to the source code, purchased commonly as off-the-shelf) ✓ Recommend a type of licence for a given scenario including benefits and drawbacks

Ethical, legal, cultural, environmental and privacy.

Issue	What it means (simple)	Example of digital technology	Impact on society
Ethical	What is right or wrong when using technology	AI facial recognition	Could reduce crime but may unfairly target groups
Legal	Laws that control how technology is used	Copyright on YouTube	People can be fined for illegal downloads
Cultural	How technology changes behaviour and lifestyle	Social media	Changes how people communicate
Environmental	How technology affects the planet	Data centres	Uses lots of electricity
Privacy	How personal data is collected and used	Smartphones tracking location	People lose control of personal information

Students lose marks in longer questions because they **list** instead of **discuss** always use **TIER** to avoid **tears**
T–I–E–R (Technology → Issues → Effects → Result)

Think in TIER. Write in paragraphs. Never list. If the question says **discuss**, aim for **6 marks** by using more than one (> 1) issue and explaining the > 1 effects.

TIER Steps	What it means	Helpful sentence starter	Example (Social Media)
T – Technology	The digital technology you are talking about	*This technology is…*	Social media
I – Issues	The problems the technology causes (more than one)	*This causes problems because… **and**…*	People share too much personal information **and** behaviour online changes
E – Effects	What happens to people or society because of the issues (more than one)	*This can lead to… and…*	People lose privacy **and** mental health or behaviour can be affected
R – Result	What happens next	*As a result…*	Privacy settings, rules and online

	because of the effects (usually one clear result)		safety awareness are used

Question (6 marks)	TIER (Thinking Structure)	Student exam answer
Discuss the ethical, cultural and privacy issues caused by the use of social media.	**T** – Social media lets people talk and share things online. **I** – This causes problems because: personal data is collected (privacy issue) AND online behaviour changes (cultural issue). **E** – This can lead to: people losing privacy AND people sharing too much information and spending less time talking face to face, which can affect mental health. **R** – As a result, privacy settings, rules and online safety awareness are used.	Social media causes privacy and cultural issues because personal data is collected and online behaviour changes. This can lead to people losing privacy and sharing too much information, as well as spending less time talking face to face, which can affect mental health. As a result, privacy settings and online safety rules are used.
Discuss the privacy and cultural issues caused by smartphones.	**T** – Smartphones are used every day for apps and messages. **I** – This causes problems because:• phones track user data (privacy issue)AND• people use phones very often (cultural issue). **E** – This can lead to:• people losing control of personal informationAND• people talking less to others in person. **R** – As a result, permission settings and screen-time controls are used.	Smartphones cause privacy and cultural issues because they track personal data and are used very often. This can lead to people losing control of their information and talking less to others face to face. As a result, phones include permission settings and screen-time controls.
Discuss the environmental and privacy issues caused by cloud storage.	**T** – Cloud storage saves data online using data centres. **I** – This causes problems because:• data centres use a lot of electricity (environmental issue)AND• personal data is stored online (privacy issue). **E** – This can lead to:• more pollution and higher energy useAND• people worrying about who can see their personal data. **R** – As a result, companies use cleaner energy and stronger security.	Cloud storage causes environmental and privacy issues because data centres use a lot of electricity and store personal data. This can lead to more pollution and make people worry about who can see their data. As a result, companies use cleaner energy and stronger security systems.

Discuss the legal and ethical issues caused by online shopping websites.	**T** – Online shopping lets people buy items on the internet. **I** – This causes problems because:• customer rights must be protected by law (legal issue)AND• personal data is collected and used (ethical issue). **E** – This can lead to:• customers being tricked or treated unfairlyAND• personal data being misused. **R** – As a result, consumer laws and secure payment systems are used.	Online shopping causes legal and ethical issues because customer rights must be protected and personal data is collected. This can lead to customers being tricked and personal data being misused. As a result, consumer protection laws and secure payment systems are used.
Discuss the ethical and privacy issues caused by artificial intelligence in everyday life.	**T** – Artificial intelligence is used to make choices and analyse data. **I** – This causes problems because:• AI decisions may be unfair (ethical issue)AND• large amounts of personal data are collected (privacy issue). **E** – This can lead to:• people being treated unfairlyAND• people losing control of their personal information. **R** – As a result, rules and controls are introduced to manage AI use.	Artificial intelligence causes ethical and privacy issues because decisions may be unfair and lots of personal data is collected. This can lead to people being treated unfairly and losing control of their information. As a result, rules and controls are introduced to manage how AI is used.

Legislation	Purpose	Allows	Prohibits
Data Protection Act 2018	Protect personal data	Lawful data processing	Misuse or insecure storage of data
Computer Misuse Act 1990	Protect systems from cybercrime	Authorised access	Hacking, malware, DoS attacks
Copyright Designs and Patents Act 1988	Protect creative work	Licensed use of content	Piracy and plagiarism

Exam Question	T – Technology	I – Issue (Law)	E – Effect (Allows / Prohibits)	R – Result (Why it matters)
Discuss the impact of the Data Protection Act 2018 on the use of online databases by organisations.	Online databases store personal data such as names, addresses, and dates of birth.	The DPA2018 applies because this is personal data.	Allows lawful data processing and secure storage but prohibits sharing data without consent or insecure storage.	Organisations must protect data properly, reducing privacy breaches and misuse.
Discuss the legal issues caused by	A school computer	The Computer Misuse Act	Allows authorised access but prohibits	Offenders can be prosecuted and

hacking into a school computer network.	network connects computers to shared files and systems.	1990 applies because hacking involves unauthorised access.	hacking, malware, and denial of service attacks.	networks are protected from disruption and damage.
Discuss the legal issues of downloading and using pirated software.	Software applications are installed and used on computers and devices.	The Copyright, Designs and Patents Act 1988 applies because software is creative work.	Allows licensed use of software but prohibits copying or downloading pirated versions.	Creators are protected and users risk legal consequences if they break the law.

Extended response – balanced discussion: A phone company makes replacement batteries difficult to obtain but offers cheaper new phones. Discuss benefits and drawbacks for the company, customers and the environment (8 marks). Use TIER to explain consequences and counterarguments, then reach a reasoned judgement; do not simply list points.

1.6.1 Exam Questions on Issues and Legislation

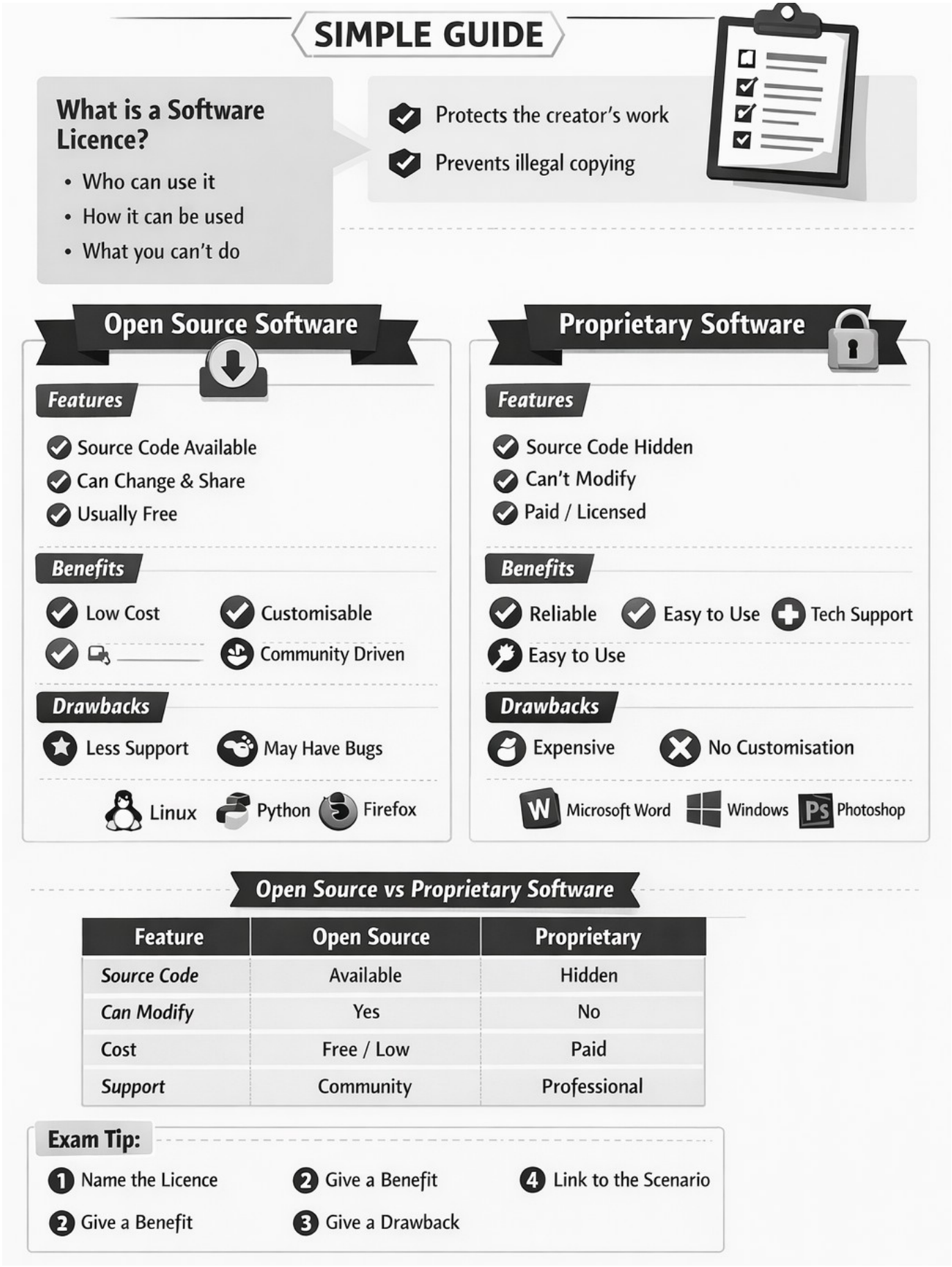
	Question type	Question	Model answer	Tip to answer (student)
1	Define	What is meant by an ethical issue in computing?	An ethical issue is about what is right or wrong when using technology.	Ethics = **right or wrong**.
1	Define	What is meant by privacy?	Privacy is how personal data is collected, stored, and used.	Think **personal data**.
1	Define	Name one law related to computer use.	Data Protection Act 2018 / Computer Misuse Act 1990 / Copyright, Designs and Patents Act 1988.	Just name one law.
2	Explain	State two social or cultural issues caused by digital technology.	Digital technology can change how people communicate and how they behave in society.	Two issues = two marks.
2	Explain	State two environmental issues caused by computer systems.	Data centres use large amounts of electricity and produce heat, increasing energy use.	Energy use + pollution.
3	Explain	Explain one privacy issue caused by smartphones.	Smartphones track location and personal data, which can cause people to lose control of their information.	Mention **tracking data**.
3	Explain	Explain one ethical issue caused by artificial intelligence.	AI systems may make biased or unfair decisions, leading to some groups being treated unfairly.	Use **unfair decisions**.
3	Explain	Explain one cultural issue caused by social media.	Social media changes how people communicate and can reduce face-to-face interaction.	Behaviour change.
4	Describe	Describe the purpose of the Data Protection Act 2018.	The Data Protection Act 2018 protects personal data by controlling how organisations collect, store, and use it.	Purpose = **protect personal data**.

4	Describe	Describe what the Computer Misuse Act 1990 protects against.	The Computer Misuse Act 1990 protects computer systems from unauthorised access, hacking, malware, and denial of service attacks.	Use **unauthorised access**.
4	Describe	Describe what the Copyright, Designs and Patents Act 1988 protects.	The Copyright, Designs and Patents Act 1988 protects creative work such as software from being copied or shared illegally.	Think **piracy**.
6	Discuss	Discuss the ethical, cultural and privacy issues caused by the use of social media.	Social media causes privacy issues because personal data is collected and shared. It also causes cultural issues by changing how people communicate and interact. These issues can lead to loss of privacy and changes in behaviour. As a result, privacy settings, rules, and online safety education are used to reduce the impact.	Use **TIER**. Write in paragraphs, not a list.
6	Discuss	Discuss the impact of the Data Protection Act 2018 on organisations using online databases.	Online databases store personal data, so the Data Protection Act 2018 applies. It allows lawful data processing but prohibits insecure storage or sharing without consent. As a result, organisations must protect data properly, reducing privacy breaches and misuse.	TIER: **Technology → Law → Effect → Result**.

1.6.2 Software Licenses

A **software licence** is a set of rules that says how software can be used and shared.

Open source software lets people see and change the code, while **proprietary software** keeps the code hidden and usually has to be paid for



1.6.2 Exam Questions on Software Licenses

M	Question type	Question	Model answer	Tip to answer (student)

1	Define	What is a software licence?	A software licence is a set of rules that says how software can be used and shared.	Licence = **rules for use**.
1	Define	State one purpose of a software licence.	To prevent illegal copying of software.	One purpose only.
1	Define	Name one type of software licence.	Open source / Proprietary.	Just name it.
2	Explain	State two purposes of a software licence.	Software licences protect the creator's work and prevent illegal copying.	Two purposes = two marks.
2	Explain	State two features of open source software.	The source code is available and users can modify the software.	Think **see code** + **change code**.
2	Explain	State two features of proprietary software.	The source code is hidden and the software is usually paid for.	Hidden code + paid.
3	Explain	Explain why open source software allows users to modify the program.	Open source software provides access to the source code, allowing users to edit and improve the software.	Mention **source code access**.
3	Explain	Explain why proprietary software is often sold as off-the-shelf software.	Proprietary software is designed for general use and cannot be modified, so it is sold ready to use.	Off-the-shelf = **ready to use**.
3	Explain	Explain one benefit of using open source software.	Open source software is often free and can be customised to suit user needs.	One benefit only.
4	Describe	Describe the difference between open source and proprietary software.	Open source software provides access to the source code and allows modification, while proprietary software hides the source code and does not allow changes.	Compare both sides.
4	Describe	Describe two drawbacks of proprietary software.	Proprietary software can be expensive and cannot be customised by the user.	Two drawbacks, one sentence each.
6	Recommend	A school wants students to learn how programs work and change the code. Recommend a type of software licence and justify your answer.	Open source software would be suitable because students can view and modify the source code, which helps learning. It is also usually free, reducing costs. However, a drawback is that professional support may be limited.	Name licence → benefits → drawback → link to scenario.
6	Recommend	A business needs reliable word-processing software with technical support. Recommend a type of software licence and justify your choice.	Proprietary software would be suitable because it is reliable, easy to use, and includes professional technical support. However, a drawback is that it can be expensive due to licensing costs.	Always include **one drawback**.
6	Discuss	Discuss the benefits and drawbacks of using open source software.	Open source software allows users to access and modify the source code and is often free, making it flexible and cost-effective. However, it may have limited technical support and can be less user-friendly.	Benefits + drawbacks = full marks.

Resources

Exam Paper 1 2026 by Topic

Question 1 – CPU, Architecture and Performance

1(a) Which component controls the operation of the CPU? **(1)**

Select one option.

☐ ALU
☐ Cache
☐ Control Unit
☐ RAM

1(b) State two registers used in the fetch–decode–execute cycle. **(2)**

1(c) Explain why registers are located inside the CPU rather than in main memory. **(3)**

1(d) Complete the table showing what each register stores: Program Counter, MAR, MDR and Accumulator. **(4)**

Program Counter: _____

MAR: _____

MDR: _____

Accumulator: _____

1(e) Explain how clock speed, cache size and number of cores together affect CPU performance. **(6)**

Question 2 – Data Representation

2(a) Binary is a number system that uses: **(1)**

Select one option.

☐ 0–9
☐ 0 and 1
☐ A–F
☐ 1–10

2(b) Explain why computers use binary to store data. **(3)**

2(c) Convert the hexadecimal number **3F** into denary.

Show your working. **(4)**

2(d) An image is 800 × 600 pixels with a colour depth of 16 bits.

Calculate the file size in MB. **(4)**

Question 3 – Memory, Storage and Embedded Systems

3(a) Match each memory type to its description: RAM, ROM and Cache. **(2)**

RAM → _____

ROM → _____

Cache → _____

3(b) Explain why embedded systems usually use ROM to store their programs. **(3)**

3(c) Explain why SSDs are often preferred to HDDs in portable devices. **(3)**

3(d) Compare RAM and ROM. **(4)**

Question 4 – Networks and Security

4(a) Which device forwards data packets between different networks? **(1)**

Select one option.

☐ Switch
☐ Router
☐ Hub
☐ NIC

4(b) Explain the role of a switch in a school network. **(3)**

4(c) Describe how encryption helps protect data when using Wi-Fi. **(4)**

4(d) Explain how data is transmitted securely from a school computer to a web server. **(6)**

Question 5 – Systems Software and Utilities

5(a) System software is software that: **(1)**

Select one option.

☐ Manages hardware and resources
☐ Creates documents
☐ Plays media
☐ Edits images

5(b) Explain one role of an operating system related to memory or process management. **(3)**

5(c) State two examples of utility software. **(2)**

5(d) Explain why utility software is important for maintaining computer performance. **(4)**

Question 6 – Ethical, Legal and Environmental Issues

6(a) State two ethical issues related to digital technology. **(2)**

6(b) Name one law related to computer use and state its purpose. **(2)**

6(c) Explain how the Data Protection Act 2018 affects how organisations store personal data. **(4)**

6(d) Discuss the impact of digital technology on individuals or society. **(8)**

End of Paper

Answers and Misconceptions by topic

Question	Right answer (student-friendly)	Common misconceptions
1(a) CPU control	**Control Unit** controls the operation of the CPU.	Choosing ALU (does calculations) or RAM (not part of CPU).
1(b) CPU registers	Any two of: **PC, MAR, MDR, Accumulator**.	Naming cache/RAM; inventing registers.
1(c) Registers location	Registers are **very fast, small storage inside the CPU** for data/addresses in use now.	Saying they store lots of data or are slower than RAM.
1(d) Register roles	**PC:** address of next instruction. **MAR:** address being accessed. **MDR:**data/instruction fetched. **Accumulator:** results of calculations.	Mixing MAR/MDR; saying PC stores data.
1(e) CPU performance	**Clock speed:** cycles per second. **Cache:** faster access to frequent data. **Cores:** parallel processing/multitasking.	Saying higher clock speed = more cores; cache replaces RAM.
2(a) Binary	Uses **0 and 1**.	Choosing A–F or 0–9.
2(b) Why binary	Hardware has **two states (on/off)** so binary is reliable.	"Computers prefer binary" with no hardware link.
2(c) Hex → denary	**3F = (3×16) + 15 = 63**.	Forgetting F=15; not showing working.
2(d) Image file size	Size = **pixels × colour depth**, then convert bits → bytes → MB.	Missing unit conversion; confusing bits/bytes.
3(a) Memory matching	**RAM:** running programs. **ROM:** start-up instructions. **Cache:** frequent data near CPU.	Saying cache is permanent; swapping RAM/ROM.
3(b) ROM in embedded	**Non-volatile**, reliable start-up every time.	Saying ROM is faster than RAM.
3(c) SSD vs HDD	**SSDs** are faster, durable, low power, no moving parts.	Saying SSDs are magnetic or have moving parts.
3(d) Compare RAM/ROM	**RAM:** volatile, used while running. **ROM:** non-volatile, stores start-up code.	Listing facts without comparison words.
4(a) Network device	**Router** forwards packets between networks.	Choosing switch (within a LAN).
4(b) Switch role	Sends frames to correct device using **MAC addresses**.	Confusing switch with router.
4(c) Encryption	**Scrambles data** so intercepted data is unreadable without a key.	Saying it speeds up Wi-Fi.
4(d) Secure transmission	Data is **encrypted**, sent via network devices, **decrypted** at destination.	Listing tools without explaining stages.
5(a) System software	**Manages hardware and resources**.	Choosing apps like documents/media.
5(b) OS role	**Memory/process management** (allocates RAM, schedules tasks).	Vague answers like "controls the computer".
5(c) Utilities	Examples: **antivirus, backup, compression, defragmentation**.	Naming apps (Word, Chrome).
5(d) Utilities importance	**Maintain**	Saying they create files for users.

	performance/security (clean, protect, organise data).	
6(a) Ethical issues	Examples: **privacy**, **addiction**, **job loss**, **surveillance**.	Naming laws instead of issues.
6(b) Law & purpose	**Data Protection Act 2018** protects personal data.	Naming a law with no purpose.
6(c) DPA impact	Data must be **secure, lawful, limited, accurate**.	Listing principles with no explanation.
6(d) Impact discussion	**Balanced**: benefits (communication, efficiency) and drawbacks (privacy, environment).	One-sided lists; no discussion.

Exam Paper 2

Question 1

1(a) Which number system is base 2? **(1)**

Select one option.

☐ Denary
☐ Binary
☐ Hexadecimal
☐ ASCII

1(b) Convert the denary number **25** into an **8-bit binary** number. **(2)**

1(c) A binary number is shifted **right by one place**.

Explain the effect this has on its denary value. **(3)**

1(d) Convert the hexadecimal number **7A** into denary.

Show your working. **(4)**

Question 2

2(a) Which register stores the address of the next instruction to be fetched? **(1)**

Select one option.

☐ MAR
☐ MDR
☐ Program Counter
☐ Accumulator

2(b) State two registers used during the fetch–decode–execute cycle. **(2)**

2(c) Explain why the Control Unit is essential in the fetch–decode–execute cycle. **(3)**

2(d) Complete the table to show what each register stores. **(4)**

MAR:	
MDR:	
Accumulator:	
Program Counter:	

Question 3

3(a) Which address uniquely identifies a device on a local network? **(1)**

Select one option.

☐ IP address
☐ MAC address
☐ URL
☐ Domain name

3(b) Give one example of a valid IPv4 address. **(1)**

3(c) State two features of a MAC address. **(2)**

3(d) Describe how a switch sends data to the correct device on a local area network. **(4)**

Question 4

4(a) Define each memory type to its correct description. **(2)**

RAM → _____

ROM → _____

Cache → _____

4(b) Explain why ROM is suitable for storing the program in an embedded system such as a washing machine. **(3)**

4(c) State two differences between RAM and ROM. **(2)**

4(d) A portable device uses an SSD rather than an HDD.

Explain two reasons why an SSD is more suitable. **(4)**

Question 5

5(a) Which operating system task controls how processor time is shared between programs? **(1)**

Select one option.

☐ File management
☐ Scheduling
☐ Compression
☐ Backup

5(b) Explain what is meant by *peripheral management* in an operating system. **(3)**

5(c) Explain why device drivers are required for peripheral devices. **(3)**

5(d) State one example of utility software and describe its purpose. **(3)**

Question 6

6(a) Which register stores the result of calculations carried out by the ALU? **(1)**

Select one option.

☐ MAR
☐ MDR
☐ Accumulator
☐ Program Counter

6(b) Add the following binary numbers.
Show your working. **(4)**

```
00110110
00011011
```

6(c) Explain how clock speed, cache memory and number of cores *together* affect CPU performance. **(6)**

Question 7

7(a) A hospital uses a wired network for desktop computers and wireless connections for tablets.

Explain one reason why **wired** connections are suitable for desktop computers in this hospital. **(3)**

7(b) Explain one reason why **wireless** connections are suitable for tablets in this hospital. **(3)**

7(c) Explain how encryption protects data when it is transmitted over a public Wi-Fi network. **(4)**

Question 8

8(a) State two ethical issues related to the use of digital technology. **(2)**

8(b) Name one law related to computer use and state its purpose. **(2)**

8(c) A student is releasing a game online.

Explain one benefit and one drawback of using an **open source licence** for this game. **(4)**

End of Paper

Answers and Misconceptions - likely OCR style exam

Question	Right answer (student-friendly)	Common misconceptions
1(a) Base 2	**Binary** uses base 2.	Choosing hexadecimal or thinking ASCII is a number system.
1(b) 25 → binary	**00011001** (16 + 8 + 1).	Missing leading zeros; adding bits instead of place values.
1(c) Right shift	Shifting right **divides the number by 2** each shift.	Saying it "moves bits" with no effect; saying it multiplies.
1(d) 7A → denary	**122** (7×16 + 10).	Forgetting A = 10; writing 7 + 10.
2(a) PC register	**Program Counter** stores the next instruction's address.	Choosing MAR or saying it stores data.
2(b) FDE registers	Any two of: **PC, MAR, MDR, Accumulator**.	Naming cache or RAM; inventing registers.
2(c) Control Unit	The CU **decodes instructions and tells components what to do**.	Saying "it controls the CPU" with no explanation.
2(d) Register roles	**MAR = address**, **MDR = data**, **PC = next instruction**, **Accumulator = results**.	Swapping MAR/MDR; saying PC stores data.
3(a) Network ID	**MAC address** uniquely identifies a device on a LAN.	Choosing IP address or URL.
3(b) IPv4 example	Any valid format e.g. **192.168.1.1**.	Numbers over 255; missing dots.
3(c) MAC features	Uses **hexadecimal**, usually **48 bits**, fixed to device.	Saying it changes location; confusing with IP.
3(d) Switch	Uses a **MAC address table** to send data only to the correct device.	Saying it broadcasts like a hub; calling it a router.
4(a) Memory match	**RAM = running programs**, **ROM = start-up**, **Cache = frequent data near CPU**.	Saying cache is permanent; swapping RAM/ROM.
4(b) ROM in embedded	ROM is **non-volatile** so the program is always there on power-up.	Saying ROM is faster than RAM.
4(c) RAM vs ROM	RAM = **volatile / temporary**, ROM = **non-volatile / permanent**.	Just listing facts with no comparison.
4(d) SSD vs HDD	SSDs are **faster, more durable, no moving parts**.	Saying SSDs are magnetic or cheaper.
5(a) Scheduling	**Scheduling** shares CPU time between programs.	Choosing file management or backup.
5(b) Peripheral management	The OS **controls peripherals using drivers**.	Writing "manages peripherals" with no detail.
5(c) Drivers	Drivers **let the OS communicate with hardware**.	Saying drivers store files or control the CPU.
5(d) Utility	e.g. **antivirus protects against malware**.	Naming apps like Word or Chrome.
6(a) Accumulator	Stores the **result of ALU calculations**.	Choosing MDR or PC.
6(b) Binary addition	Correct binary sum with **working shown**.	Adding like denary; missing carry bits.
6(c) CPU performance	**Clock speed = cycles**, **cache = faster access**, **cores = parallel processing**.	Saying "faster CPU" repeatedly; mixing factors up.
7(a) Wired (hospital)	**Reliable, fast, secure** for fixed desktop use.	Giving generic advantages with no hospital link.
7(b) Wireless (tablets)	**Mobility and flexibility** for moving staff.	Saying "it's better" with no reason.
7(c) Encryption	Data is **scrambled** so intercepted data is	Saying encryption blocks

	unreadable without the key.	hackers directly.
8(a) Ethical issues	e.g. **privacy, addiction, surveillance, job loss**.	Naming laws instead of issues.
8(b) Law & purpose	e.g. **Data Protection Act – protects personal data**.	Naming a law with no purpose.
8(c) Open source licence	**Benefit:** can edit/share code. **Drawback:** less control/profit.	Only listing benefits; not linking to game scenario.

More Resources

Revision QR Poster – Scan to Revise



OCR Official Past Papers & Examiner Reports



Craig 'n' Dave – OCR J277 Videos



BBC Bitesize – OCR GCSE CS



Isaac Computer Science – OCR GCSE



Physics & Maths Tutor – OCR GCSE



Seneca Learning – GCSE CS